

V1 Product Concept Document

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Purpose of This Document

This document serves as the foundational product specification for the Version 1 release of the Hostel Management Platform. It is authored as a CTO-level design blueprint, structured to guide the product, engineering, and design teams through the full scope of V1. It covers every feature module, user journey, data architecture, infrastructure decisions, and the reasoning behind every major design choice.

This document is the source of truth before the Product Requirements Document (PRD) is authored. All subsequent engineering tickets, design specs, and API contracts must trace back to this document.

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1. Executive Summary

The Hostel Management Platform began as a Minimum Viable Product (MVP) designed to digitize the core operations of a multi-hostel PG management business: tenant onboarding, room allocation, basic rent tracking, and food billing. The MVP validated the core hypothesis — that hostel operators in India are deeply underserved by generic property management tools and need a purpose-built solution.

Version 1 is not an incremental improvement. It is a fundamental architectural and product evolution.

V1 transforms the platform from a **digital ledger** — a system that records what happened — into a **Hostel Operating System (HOS)**: a comprehensive, intelligence-driven platform that manages every operational dimension of running a hostel business at scale.

What V1 Introduces

- A QR-code-driven, self-service tenant onboarding funnel with real-time bed locking
- A sophisticated multi-mode billing engine with normalization, proration, tax groups, and late fees
- A complete food management system with per-meal pricing, food wallets, procurement reporting, and configurable cancellation windows
- Smart bed operations — transfer, swap, reservation, and waiting list management
- A legally-compliant digital Stay Pass with automated lifecycle management
- An employee attendance system with geolocation verification and leave management
- A multi-metric hostel performance scoring engine with transparent sub-scores
- A Finance Center with invoicing, receipts, expense governance, and bill vault
- DPDP Act 2023 compliance infrastructure
- A complete Visitor and Guest log for security and compliance
- Social login (Google, Facebook) for the tenant portal

What V1 Explicitly Defers to V1.1

- Canvas/drag-and-drop dashboard widget system
- Regional language assistant (Malayalam)

Infrastructure Commitment

V1 operates on the AWS Free Tier (ap-south-1, Mumbai) with a target infrastructure cost of under Rs.500/month for the first 12 months.

2. MVP to V1: The Evolution

2.1 What the MVP Established

The MVP successfully delivered the foundational scaffolding of the platform:

CAPABILITY	MVP STATUS
Multi-tenant organizational structure	Complete
Hostel / Floor / Flat / Room / Bed hierarchy	Complete
Warden-initiated onboarding with admin approval	Complete
Tenant portal with rent and stay visibility	Complete
Basic payment proof upload (UPI screenshot)	Complete
Food billing system (admin-managed)	Complete
KYC document upload (Aadhaar, PAN)	Complete
Activity logs and audit trail	Complete
Basic task assignment for wardens	Partial
In-app notifications	Partial
AWS deployment (EC2 + RDS + S3 + CloudFront)	Complete

2.2 The MVP's Structural Limitations

While the MVP proved the concept, it has identifiable gaps that prevent deployment to real customers at scale:

- 1. Onboarding is warden-dependent.** Every tenant onboarding requires a warden to initiate the flow and send a link manually. There is no self-service mechanism where a prospective tenant can scan a QR code, see available rooms, and begin the booking process independently.
- 2. Billing is manual.** Rent amounts are entered manually per tenant with no engine to automatically calculate prorated bills, apply normalization cycles, or compute late fees. This does not scale past 20 tenants.
- 3. Food is not tenant-driven.** Tenants cannot manage their own food plan, book specific meals, or view what is being served. The food system is entirely admin-managed with no procurement intelligence.
- 4. No financial intelligence.** The platform cannot tell the admin how much the business made last month, what the expenses were, which category of expense grew, or what next month's expected revenue is.

- 5. No employee operations.** There is no system for tracking hostel staff — their attendance, their tasks, their leave, or their operational expenses.
- 6. No bed lifecycle management.** Once a tenant checks into a bed, there is no workflow to transfer, swap, or reserve beds. Bed history for audit purposes is incomplete.
- 7. No legal compliance.** The platform stores Aadhaar and PAN data without explicit DPDP consent workflows or document deletion mechanisms — a legal liability for any commercial deployment.

2.3 V1 Design Principles

V1 is built around five non-negotiable principles:

PRINCIPLE	WHAT IT MEANS
Self-service over manual	Every tenant interaction that requires warden intervention should have a self-service option
Financial clarity	Every rupee in and out of every hostel should be visible, categorized, and queryable in real time
Operational completeness	A warden should run their entire hostel from this platform without any other tool
Compliance by design	Privacy, consent, and audit requirements are built into the core data model, not added as an afterthought
Scale without re-architecture	V1 is designed to handle 10x growth from the MVP without requiring fundamental re-architecture

3. Product Philosophy & Design Direction

3.1 Visual Design Language

V1 introduces a complete visual redesign. The design system is inspired by modern fintech dashboard aesthetics — specifically the Pinpoint/Finponin design language — characterized by:

- **Primary Accent:** Rich, saturated orange (approximately HSL 28, 95%, 55%) used for key metrics, primary CTAs, and performance indicators
- **Neutral Base:** Near-black dark grays (`#111827` , `#1F2937`) for backgrounds and data containers
- **Positive/Negative Indicators:** Orange for revenue and positive outcomes; muted red for deficits, overdue items, and urgent states
- **Typography:** Plus Jakarta Sans or Inter — clean, modern, with strong numerical weight for financial figures
- **Card Architecture:** Bento-box style card grid layouts with subtle elevation. No heavy drop shadows.
- **Fixed Dashboard Layout (V1):** A carefully designed, fixed widget layout. The drag-and-drop canvas is a V1.1 feature.

3.2 Mobile–First for Tenants and Employees

The tenant portal and QR booking flow must be optimized for mobile browsers. All tenant-facing and employee-facing screens must be designed mobile-first because:

- Prospective tenants scan QR codes on their phones in the field
- Tenants access their Stay Pass, food calendar, and payment status from their phones
- Employees mark attendance on their phones at the hostel premises

3.3 Desktop–First for Operations

The admin dashboard, warden panel, comparisons module, expense management, and reporting are primarily used on desktop browsers. Responsive mobile support is required but the layout priority is desktop.

3.4 Design Reference

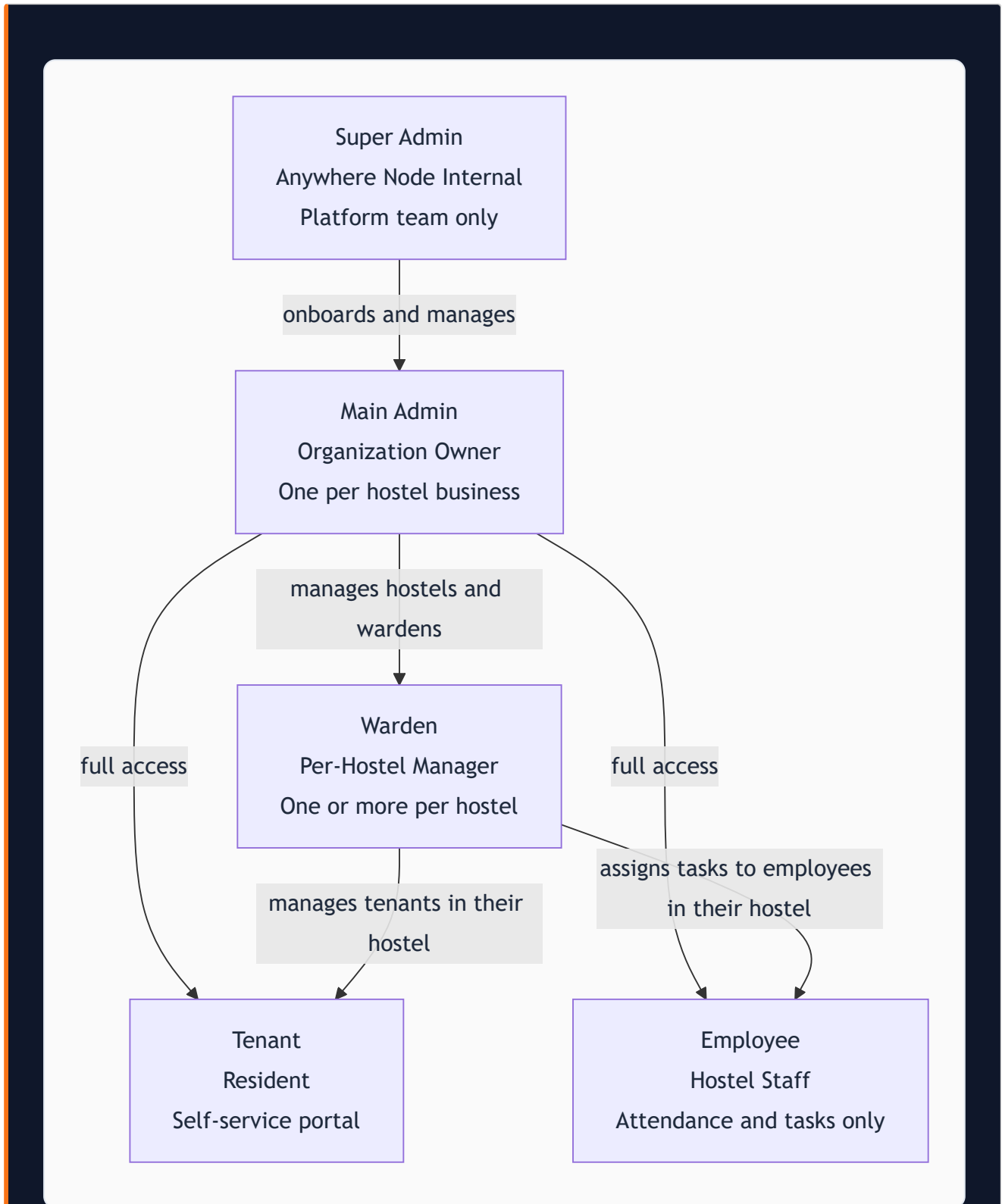
The left navigation pane, color usage, and financial metric card layout follows the Pinpoint/Finponin reference design:

- **Left pane:** Dark background, orange accents on active items, icon + label per module

- **Top bar:** Contextual to the current page — dropdowns and actions relevant to what the user is viewing
 - **Main content:** Card-based, data-dense but not cluttered
 - **Financial figures:** Large, bold, prominently placed with MoM trend indicators
-

4. User Roles & Permission Architecture

4.1 Role Hierarchy



4.2 Role Definitions

Super Admin — Anywhere Node Internal

The highest privilege tier, used exclusively by the Anywhere Node platform team. Never exposed to hostel operators.

Capabilities:

- Onboard new organizations and create Main Admin accounts
- Manage platform-wide settings and feature flags per organization
- Access any organization for support and debugging
- Monitor platform-wide health and usage metrics

Main Admin — Organization Owner

One per organization (e.g., a PG business with 5 hostels has one Main Admin). Owns all configuration decisions.

Capabilities:

- Full read/write access to all hostels, wardens, tenants, and employees in their organization
- Configure all financial settings: UPI ID, QR, tax groups, currency display, normalization, late fees
- Create and manage expense entities and expense tags
- Approve or reject all warden-submitted expenses and expense vouchers
- Approve or reject employee leave requests (exclusive to Main Admin)
- Access all analytics, comparisons, MRR reports, and financial statements
- Generate and review all invoices, receipts, and expense vouchers
- Create and maintain the weekly food menu and food pricing
- Access the DPDP data deletion panel

Warden — Per-Hostel Manager

One or more wardens per hostel. The operational layer between admin and tenants.

Capabilities:

- Manage onboarding, bed assignment, and tenant lifecycle within their hostel
- Review and verify tenant payment proofs
- Generate bills and issue Stay Passes
- Apply discounts to bills

- Log expenses (only for expense entities/tags they have been granted access to)
- Access their hostel's food procurement report
- Manage the visitor/guest log for their hostel
- Assign tasks to employees in their hostel
- Schedule housekeeping for their hostel
- Send announcements to tenants in their hostel
- Toggle bill normalization for their own hostel only
- Create expense voucher requests

Tenant — Resident

The end resident. Access is scoped entirely to their own data.

Capabilities:

- View their own stay details, rent, and payment history
- Manage their food plan, food wallet, and meal bookings
- Submit service requests (maintenance requests, complaints)
- View, download, and share their Stay Pass
- Request stay renewal or extension
- View and respond to announcements

Employee — Hostel Staff

Hostel operational staff. Minimal, task-focused access.

Capabilities:

- Mark daily attendance (geolocation-verified)
- Mark daily checkout (geolocation-verified)
- Submit leave requests
- View their own attendance history and working hours
- View and update tasks assigned to them

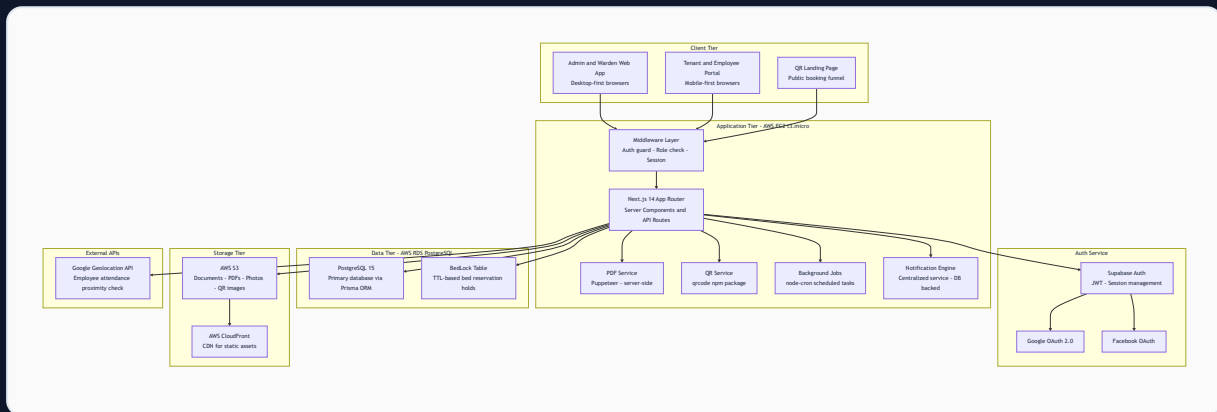
4.3 Permission Matrix

FEATURE	SUPER ADMIN	MAIN ADMIN	WARDEN	TENANT	EMPLOYEE
Manage organizations	Yes	-	-	-	-
View all hostels	Yes	Yes	Own only	-	-
Create warden accounts	Yes	Yes	-	-	-
Create tenant accounts	Yes	Yes	Yes	-	-
Create employee accounts	Yes	Yes	Yes (own hostel)	-	-
Onboard tenant (initiate)	Yes	Yes	Yes	Self via QR	-
Approve tenant onboarding	Yes	Yes	Yes	-	-
Generate bills	Yes	Yes	Yes	-	-
Apply discounts to bills	Yes	Yes	Yes	-	-
Verify payments	Yes	Yes	Yes	-	-
Issue Stay Pass	Yes	Yes	Yes	-	-
Configure UPI QR	Yes	Yes	-	-	-
Create and manage tax groups	Yes	Yes	-	-	-
Toggle bill normalization	Yes	Yes (all hostels)	Yes (own hostel)	-	-
Configure late fee policy	Yes	Yes	-	-	-
Create expense entities	Yes	Yes	-	-	-
Create expense tags	Yes	Yes	-	-	-
Log expenses	Yes	Yes	Yes (by entity access)	-	-
Approve expenses	Yes	Yes	-	-	-
Create expense vouchers	-	Yes	Yes	-	-
Approve expense vouchers	Yes	Yes	-	-	-

FEATURE	SUPER ADMIN	MAIN ADMIN	WARDEN	TENANT	EMPLOYEE
View food procurement report	Yes	Yes	Yes (own hostel)	-	-
Set weekly food menu	Yes	Yes	-	-	-
Manage food pricing	Yes	Yes	-	-	-
Book and manage meals	-	-	-	Yes	-
Access comparisons module	Yes	Yes	-	-	-
View MRR report	Yes	Yes	-	-	-
Mark attendance	-	-	-	-	Yes (own)
Request leave	-	-	-	-	Yes (own)
Approve leave	Yes	Yes	-	-	-
Assign tasks to employees	Yes	Yes	Yes (own hostel)	-	-
View visitor log	Yes	Yes	Yes (own hostel)	-	-
Log visitor entry and exit	Yes	Yes	Yes	-	-
Submit service requests	-	Yes	Yes (incidents)	Yes (complaints/maintenance)	-
Resolve service requests	Yes	Yes	Yes	-	-
Delete KYC documents (DPDP)	Yes	Yes	-	-	-
Generate comparison reports	Yes	Yes	-	-	-
Configure geofence radius	Yes	Yes	-	-	-
Manage asset register	Yes	Yes	Yes (own hostel)	-	-
Schedule housekeeping	Yes	Yes	Yes (own hostel)	-	-
Send announcements	Yes	Yes (all hostels)	Yes (own hostel)	-	-

5. System Architecture Overview

5.1 High-Level Architecture



5.2 AWS Services — V1 Scope

SERVICE	PURPOSE	COST TIER
EC2 t3.micro	Next.js application server (Dockerized)	Free Tier - 750 hrs/month
RDS PostgreSQL t3.micro	Primary relational database	Free Tier - 750 hrs/month
S3	Document, photo, PDF, and QR image storage	Free Tier - 5 GB
CloudFront	CDN for static assets and media delivery	Free Tier - 1 TB transfer
ACM	SSL certificate for custom domain	Free
SSM Parameter Store	Secrets management (DB URL, API keys)	Free Tier
CloudWatch	Application logs, metrics, alarms	Free Tier - 5 GB
IAM	Role-based AWS resource access control	Free

Infrastructure Cost Target: Under Rs.500/month for the first 12 months of operation.

5.3 Core Technical Stack

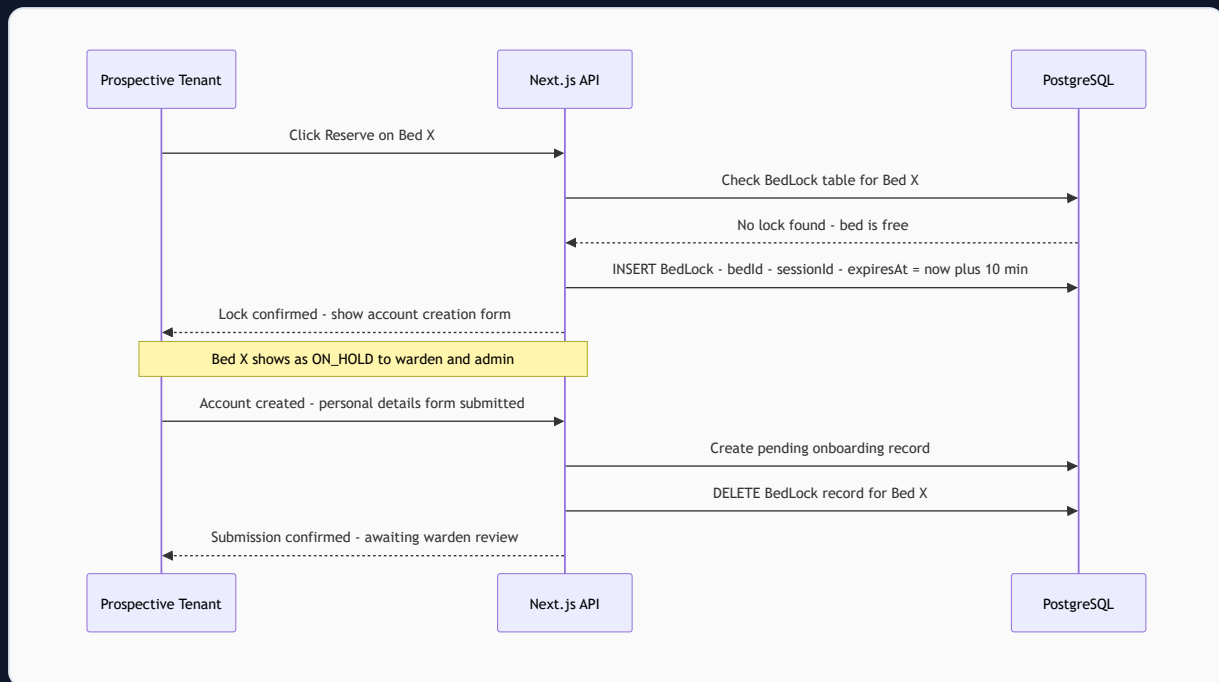
LAYER	TECHNOLOGY	RATIONALE
Framework	Next.js 14 App Router	Full-stack SSR, API routes, Server Components in one package
Database ORM	Prisma	Type-safe queries, schema migrations, excellent developer experience
Database	PostgreSQL 15 via AWS RDS	ACID compliance, relational integrity for financial data
Auth	Supabase Auth	Managed auth, JWT, OAuth providers, Row Level Security
File Storage	AWS S3 with pre-signed URLs	Secure, cost-effective, no permanent public URLs for sensitive docs
PDF Generation	Puppeteer (headless Chromium, server-side)	Pixel-perfect HTML-to-PDF, full design control
QR Generation	qrcode npm package	Server-side generation, no third-party dependency
Real-Time Data	SWR polling at 5-second interval	Zero infrastructure overhead, smart pausing when tab is hidden
Geolocation	Browser Geolocation API + Google Maps SDK	Proximity verification for employee attendance
Language	TypeScript end-to-end	Type safety across frontend, backend, and database layer
Background Jobs	node-cron inside Next.js process	No separate infrastructure, minimal complexity

5.4 Real-Time Strategy

V1 uses **SWR polling every 5 seconds** for real-time dashboard updates — a deliberate architectural choice detailed in SDD v1.7. This applies to: admin dashboard financial metrics, new onboarding requests, pending payment queue, expense approval queue, service request status, bed availability counts, and food order counts.

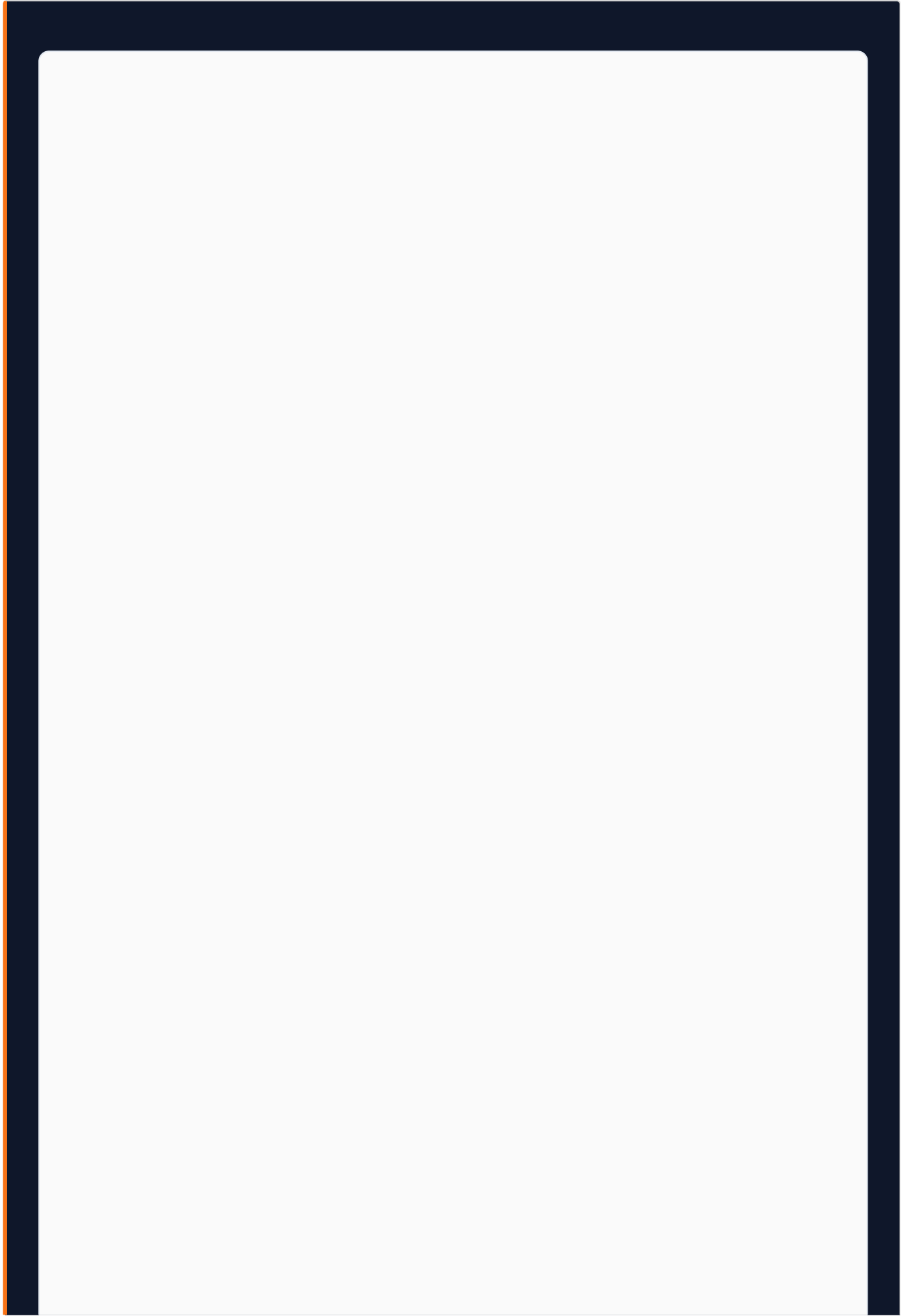
SWR pauses automatically when the browser tab is backgrounded. Server-Sent Events (SSE) is the planned Phase 2 upgrade when real-time requirements become more demanding at scale.

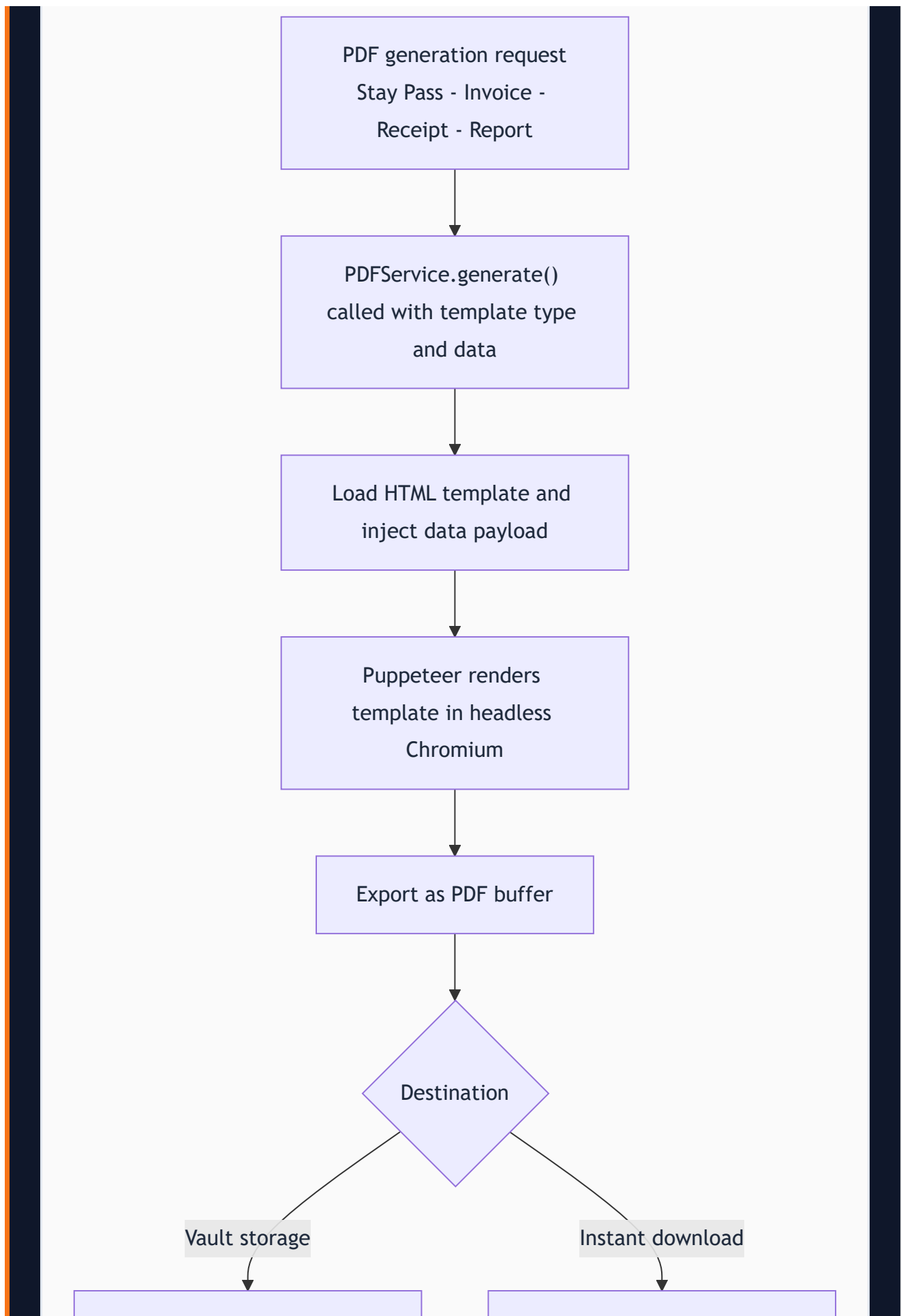
5.5 Bed Locking Architecture

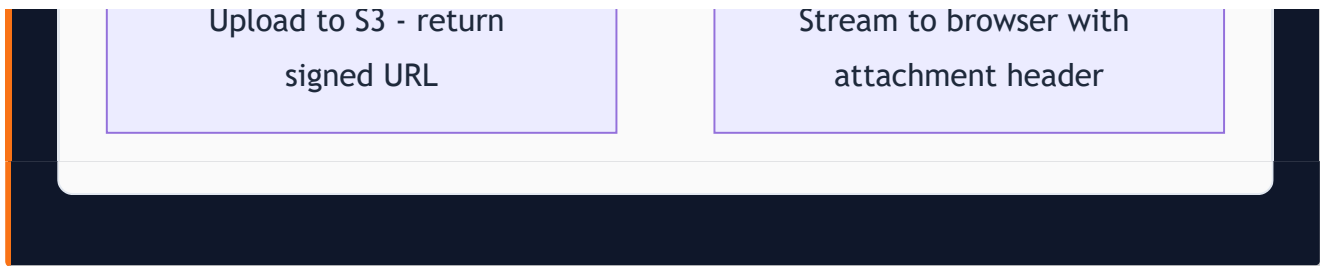


A node-cron job runs every 5 minutes to clean expired **BedLock** records, automatically restoring abandoned beds to **AVAILABLE** status.

5.6 Centralized PDF Generation Service

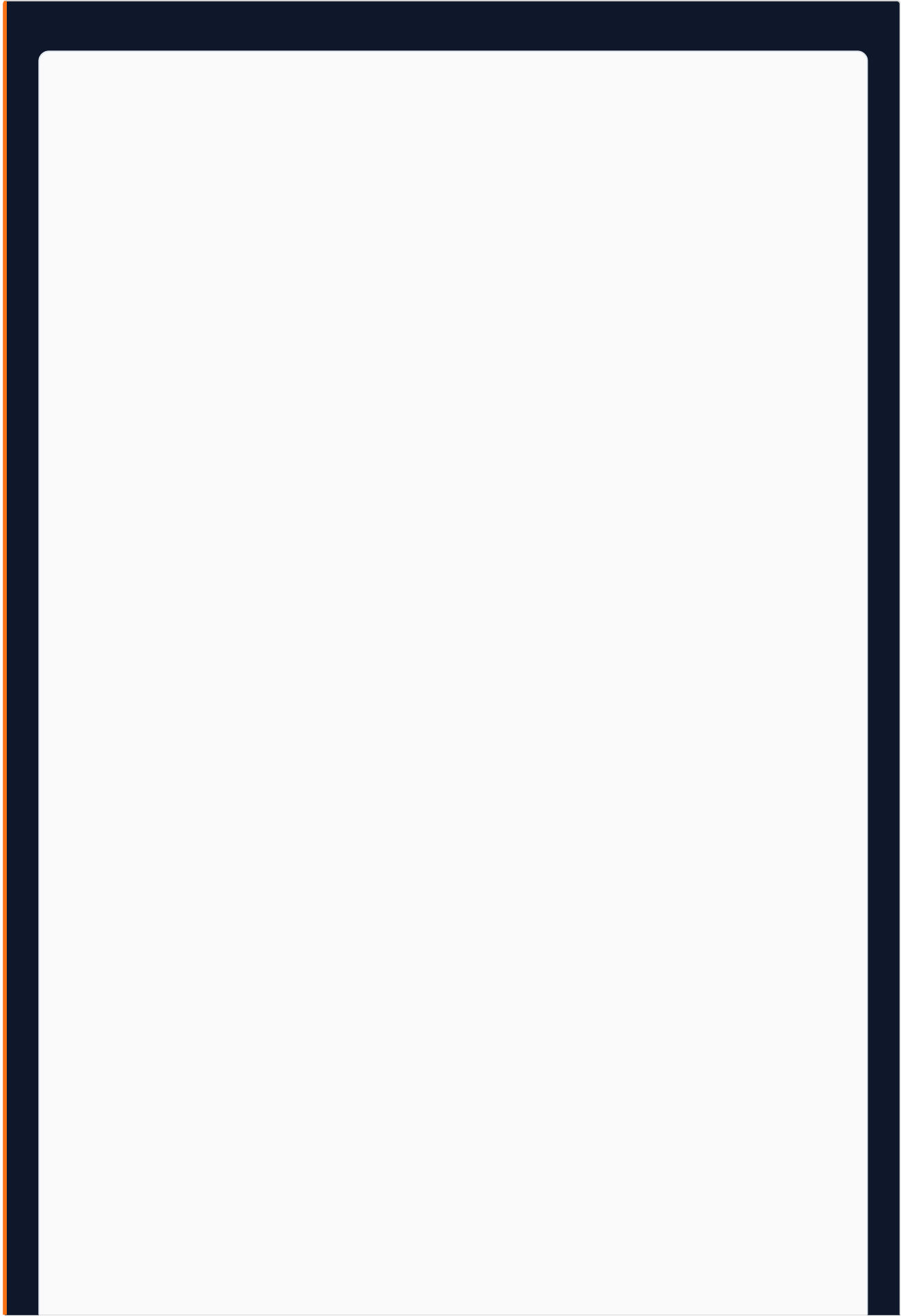


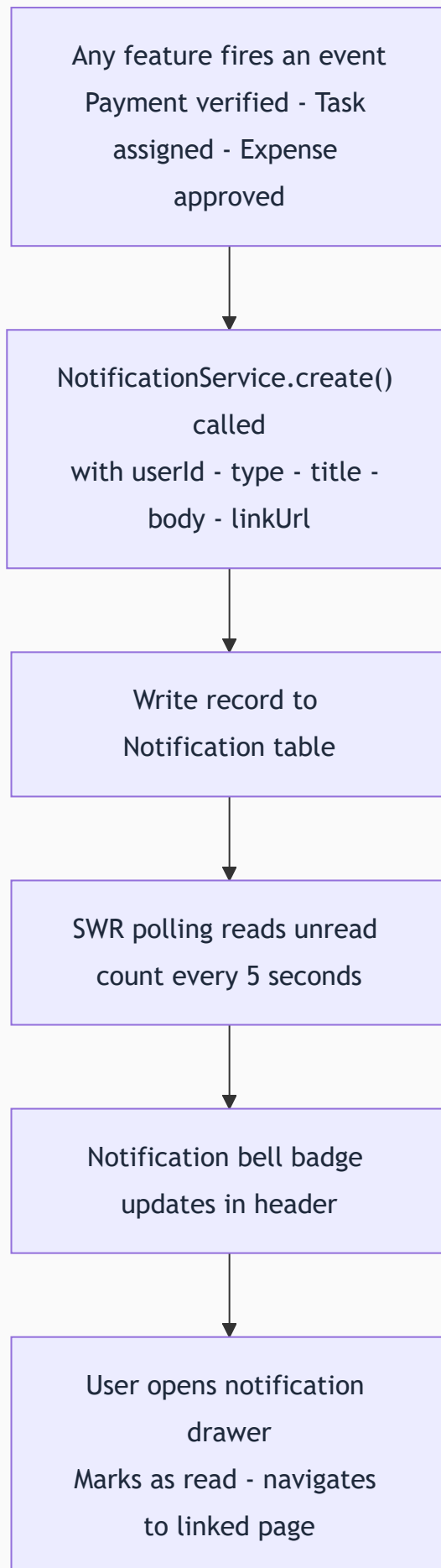




All PDFs share consistent branding, headers, footers, and page formatting. No feature builds ad-hoc PDF logic.

5.7 Centralized Notification Engine





V1 Notification Types:

TYPE	TRIGGERED BY	RECIPIENTS
PAYMENT_VERIFIED	Warden verifies payment	Tenant
BILL_GENERATED	Bill created after onboarding approval	Tenant
STAY_EXPIRING	Cron - X days before stay end	Tenant and Warden
STAY_PASS_READY	Stay Pass generated	Tenant
ONBOARDING_SUBMITTED	Prospective tenant submits form	Warden and Admin
EXPENSE_PENDING	Warden submits expense	Admin
EXPENSE_APPROVED	Admin approves expense	Warden
VOUCHER_APPROVED	Admin approves expense voucher	Warden
TASK_ASSIGNED	Admin or Warden assigns task	Employee
LEAVE_APPROVED	Admin approves leave	Employee
LEAVE_REJECTED	Admin rejects leave	Employee
SERVICE_REQUEST_UPDATED	Warden updates status	Tenant
BED_AVAILABLE	Cron - bed freed while tenant on waitlist	Warden to contact
FOOD_CUTOFF_WARNING	Cron - 1 hour before cutoff	Tenants with unbooked meals
FOOD_WALLET_LOW	Food booking - wallet below threshold	Tenant

5.8 Integration-Ready API Design

V1 does not integrate with any third-party tools. However, the APIs are designed for future integrations without refactoring:

- 1. API Key Infrastructure:** The `Organization` model contains an auto-generated `apiKey` (UUID). Future integrations authenticate using this key.
- 2. Webhook Architecture Foundation:** `WebhookEndpoint` and `WebhookLog` models are defined in the schema. `webhookService.emit(event, payload)` is a no-op in V1 but becomes a real HTTP call when integrations are enabled.
- 3. Structured Event Naming:** All internal events follow `resource.action` convention:
`tenant.onboarded` , `payment.verified` , `expense.approved` , `stay_pass.issued` .

6. Module 01: Dashboard & Analytics

6.1 Overview

The main dashboard is the command center for the Main Admin, providing a real-time financial and operational overview of the entire organization. V1 ships with a carefully designed fixed layout.

6.2 Main Admin Dashboard Layout

Section A — Organization Financial Summary (Top Row)

Three headline metrics for the current month:

- **Total Sales** (highlighted prominently — the primary gross revenue indicator)
- **Total Expenses** (broken down by category on hover)
- **Net Revenue / Net Profit** = Total Sales minus Total Expenses (displayed in orange — the primary brand metric)

Month-over-Month (MoM) trend indicator beside each metric (percentage up or down vs previous month).

Section B — Expense Breakdown Cards

Utility-icon cards, one per expense category, showing current month total:

- WiFi, Electricity (lightning bolt icon), Water (droplet), Housekeeping (broom), Building Rent (building), Food/Beverages (fork icon), Others (tag)

Section C — Facility Selector

Dropdown in the top bar: "All Facilities" by default. Selecting a specific hostel filters all dashboard metrics to that hostel only.

Section D — Hostel Allocation Performance Grid

Grid of hostel performance cards (Pinpoint/Finponin bento-box style). Each card shows:

- Hostel name, revenue contribution as % of total, occupancy rate, Performance Score badge (Green/Amber/Red), quick stats

Clicking any hostel card navigates to that hostel's isolated dashboard.

Section E — Insights Panel

System-generated operational observations:

- "NextHome Paradise occupancy dropped 15% in the last 7 days"

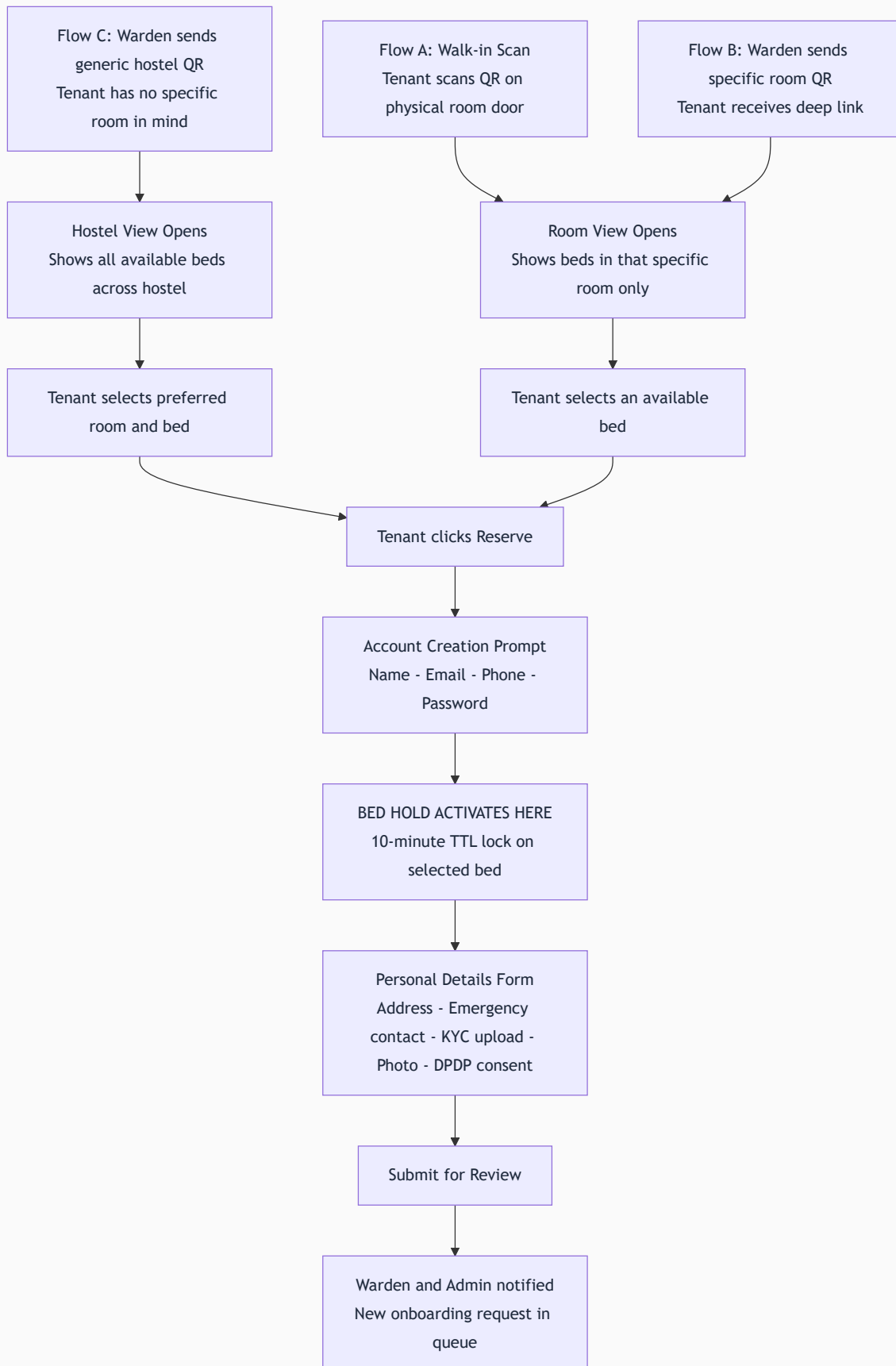
- "3 tenants at NHC have payments overdue by 7+ days"
- "Food plan adoption at NHP is below 30% this month"

6.3 Financial Metric Definitions

METRIC	FORMULA	SCOPE
Gross Rent Revenue	Sum of all verified rent payments	Per hostel or org
Food Revenue	Sum of all food wallet top-ups verified	Per hostel or org
Total Revenue	Gross Rent plus Food Revenue	Per hostel or org
Total Expenses	Sum of all admin-approved expenses	Per hostel or org
Net Profit	Total Revenue minus Total Expenses	Per hostel or org
Occupancy Rate	$(\text{Occupied Beds} / \text{Total Beds}) \times 100\%$	Per hostel
Food Adoption	$(\text{Tenants with active food plan} / \text{Total}) \times 100\%$	Per hostel

7. Module 02: Booking & Onboarding System

7.1 The Three Booking Flows



Privacy Rule: QR URLs encode a signed JWT token scoped to a specific room (Flows A/B) or specific hostel (Flow C). A prospective tenant cannot browse beyond what the QR authorizes.

7.2 Bed Selection UI

A mobile-optimized page showing:

- Hostel name and logo
- Room identifier
- Visual bed layout with color-coded status: Green (Available), Gray (Occupied), Yellow (On Hold), Blue (Reserved)
- Tapping a bed: slide-up sheet with details and "Reserve This Bed" CTA

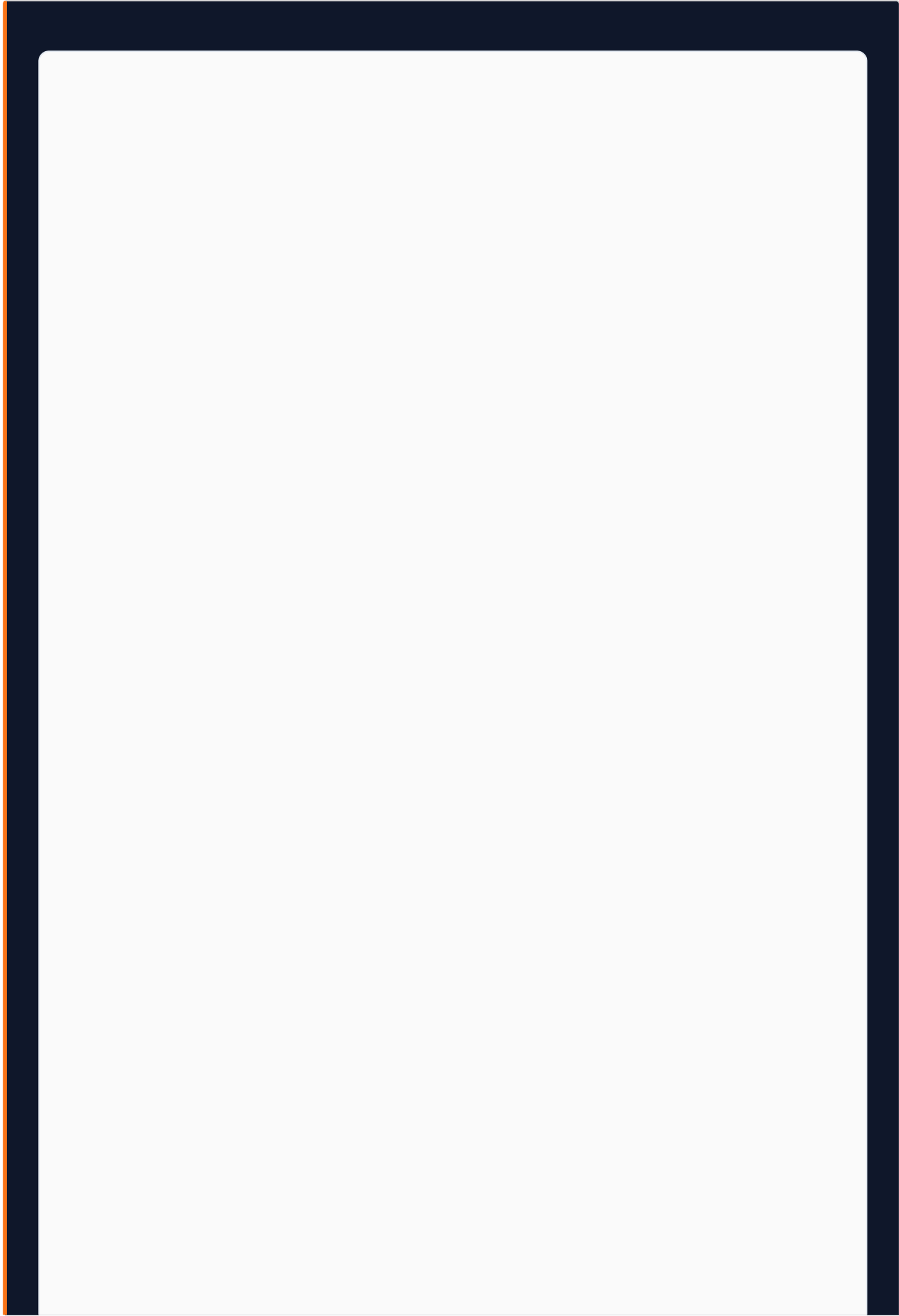
7.3 Bed Hold Activation

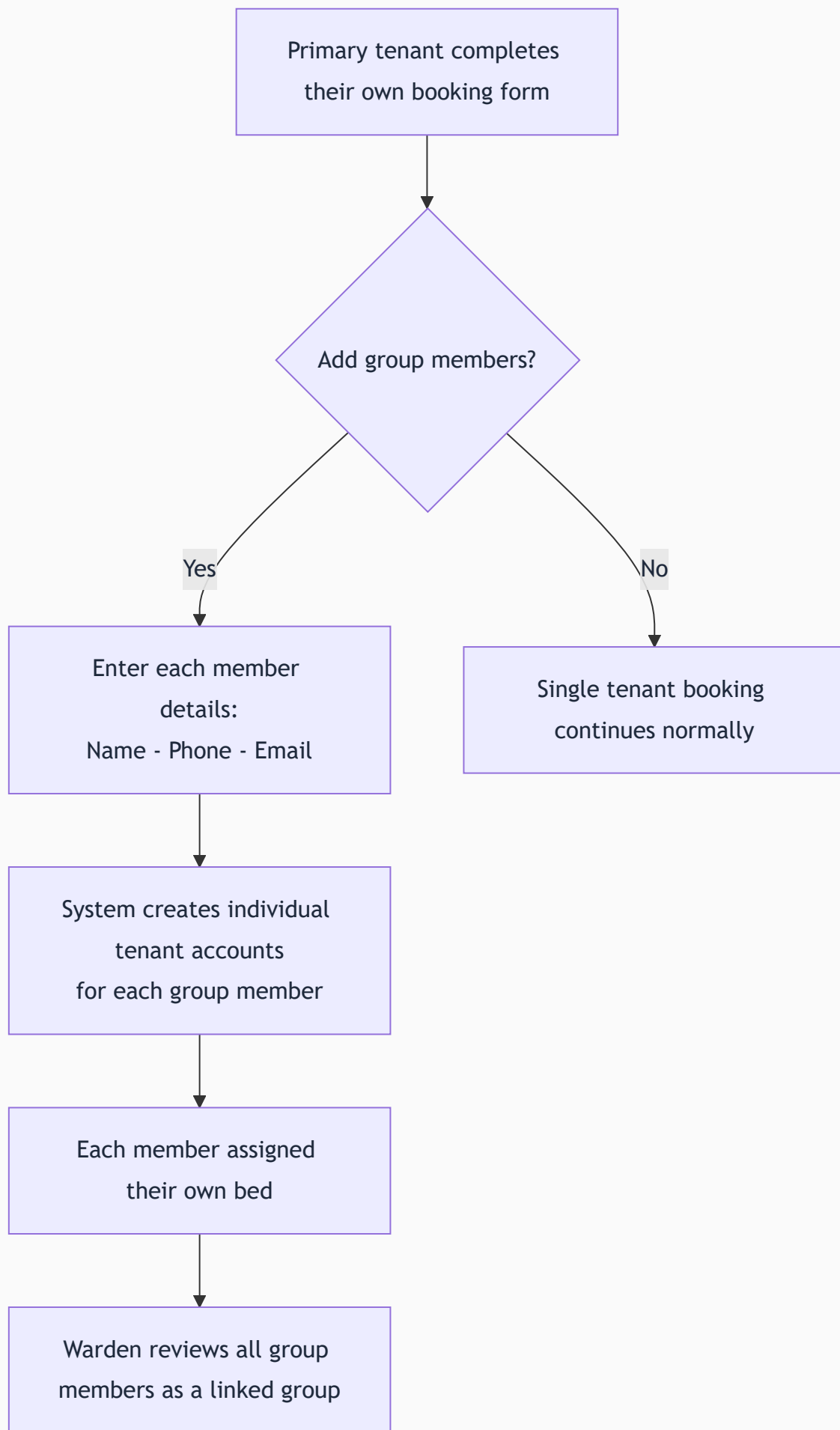
The bed hold activates at the exact moment the prospective tenant creates their account (not at the "Reserve" click). This is the right balance — we confirm interest (account creation) before locking the bed.

7.4 Personal Details Form

FIELD	REQUIRED	NOTES
Date of Birth	Yes	
Permanent Address	Yes	Full address with pin code
Emergency Contact Name	Yes	
Emergency Contact Phone	Yes	
KYC Document Type	Yes	Aadhaar / PAN / Driving License
KYC Document Number	Yes	
KYC Document Scan	Yes	Photo upload to S3
Tenant Photo	Yes	Photo or camera capture to S3
DPDP Consent Checkbox	Yes	With timestamp recorded server-side

7.5 Group Booking Flow





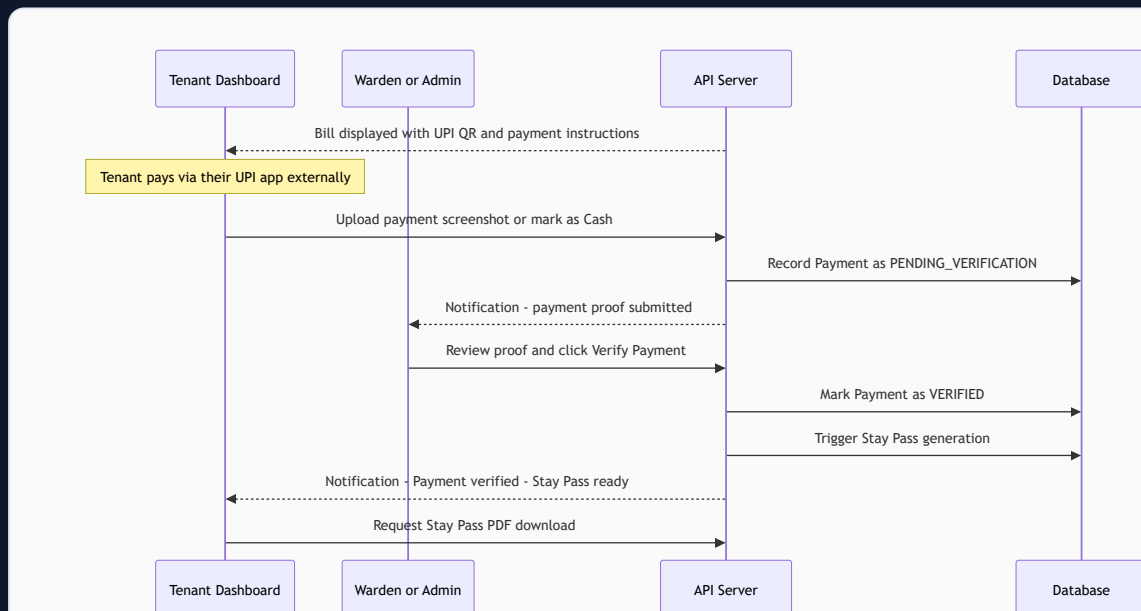
Individual Stay Passes
generated for each
member after payment

7.6 Warden Review and Bill Creation

After submission, warden/admin sees:

1. **Details Tab:** All personal information submitted
2. **Documents Tab:** KYC scan and tenant photo with zoom
3. **Bill Creation Tab:** Select rent type, enter amount, apply tax group, apply discount, assign Admission ID
4. **Action:** Click "Verified and Create Bill"

7.7 Payment Verification Flow



8. Module 03: Billing Engine

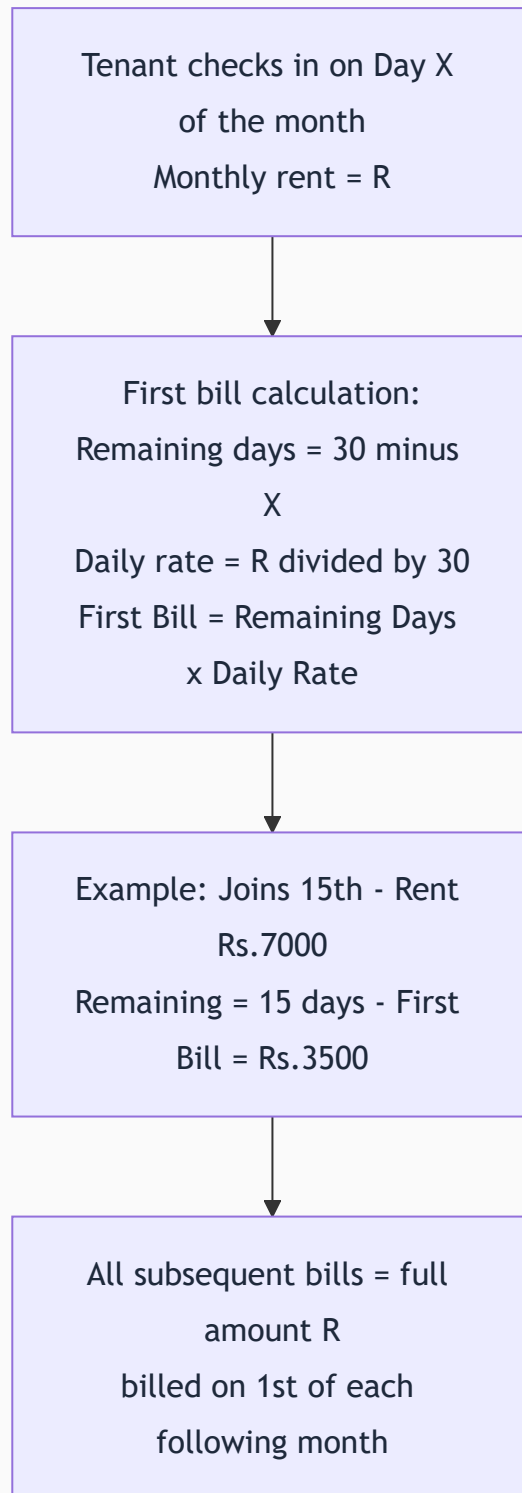
8.1 Rent Types

TYPE	DESCRIPTION	NORMALIZATION COMPATIBLE
Daily	Rent calculated per day	No
Weekly	Rent per 7-day period	No
Monthly	Rent per 30-day period	Yes

8.2 Bill Normalization

Aligns all monthly-rate tenant billing cycles to the 1st of each calendar month.

Control: Main Admin can enable for any hostel. Wardens can enable only for their own hostel.



Normalization applies only to monthly-rate tenants. Daily and weekly tenants are not subject to normalization.

8.3 Discount Application

At any bill creation step, warden or admin can apply:

- Fixed amount (e.g., Rs.500 off)
- Percentage (e.g., 10% off)

Discount reason is required for audit trail. Visible on invoice and receipt.

8.4 Tax Groups

Created by Main Admin with: name, tax percentage, applicability (rent/food/both). Applied at bill creation. Invoice shows: Base Amount, Tax Amount, Total Payable.

8.5 Late Payment Fees

When enabled by Main Admin:

- Grace period (days after due date before fee applies)
- Fee type: Fixed amount OR percentage of overdue amount
- A cron job runs on the 1st of each month for normalized hostels, appending late fees to overdue bills
- Only available on hostels with normalization enabled (predictable due dates required)

8.6 Advance Rent Refunds

When a tenant leaves before their paid period ends:

1. Admin/Warden opens tenant record, clicks "Issue Refund"
2. System shows: days remaining and calculated refund amount
3. Admin/Warden enters actual refund amount (may differ — e.g., deductions)
4. Refund reason required for audit
5. Recorded in tenant payment history and org financial records

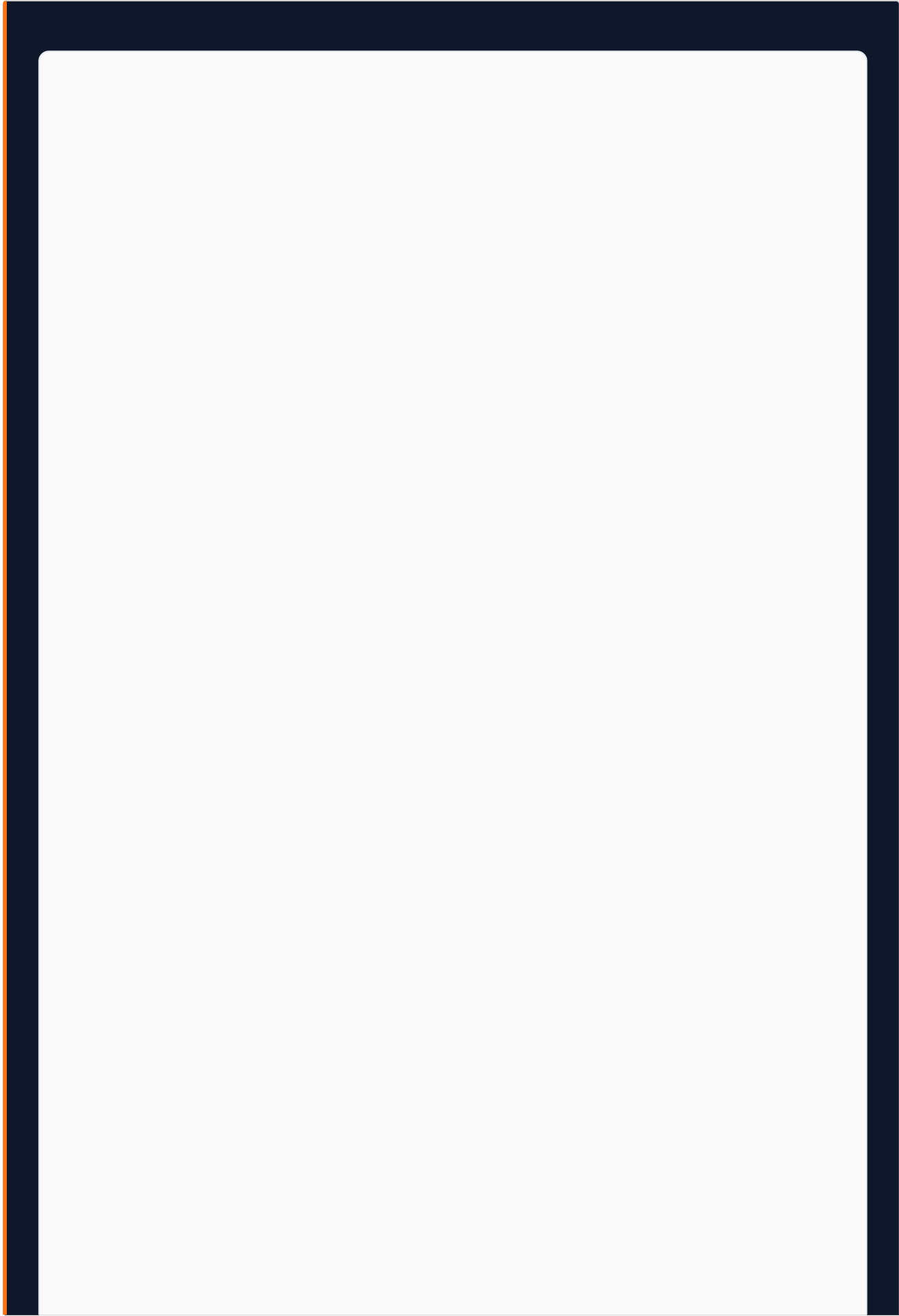
8.7 Stay Extension

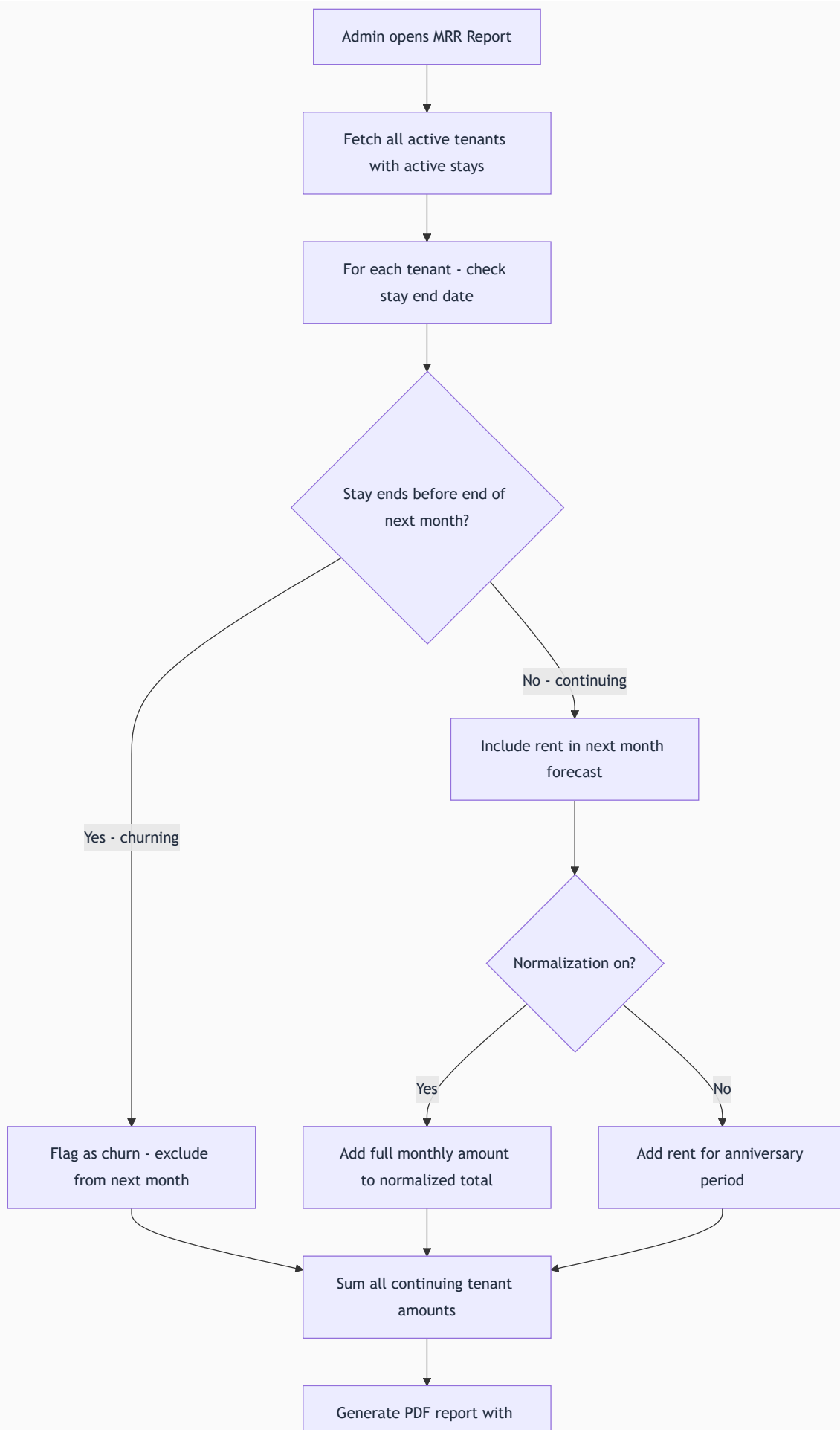
Warden/Admin selects extension type:

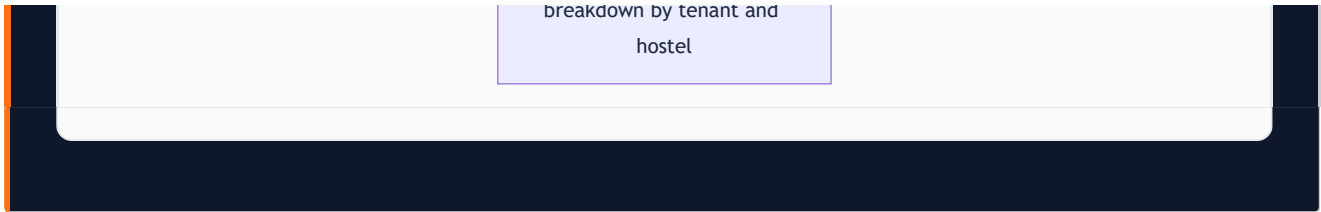
- Add X more days (at daily rate)
- Add X more weeks (at weekly rate)
- Switch to monthly

System generates extension bill. Standard payment flow initiates. New Stay Pass generated after payment.

8.8 Next Month MRR Report







9. Module 04: Finance Center

9.1 Three Financial Documents

Invoice (Pre-Payment Bill): Generated after onboarding verification. Contains: invoice number (sequential), invoice date, due date, tenant details, room/bed, billing period, base rent, tax, discount, total payable, UPI QR code and UPI ID, payment instructions.

Tenant Payment Receipt: Generated after payment is verified. Contains: receipt number, verification date and time, tenant details, amount paid, payment method (UPI/Cash), billing period covered, "PAID IN FULL" stamp.

Both are downloadable as PDF by the tenant and stored in the Bill Vault.

Expense Voucher (Petty Cash): Warden-initiated cash request for operational expenses. Contains: voucher number, purpose, amount, expense tag, attachment, approval timestamp, approved by (admin name).

Vouchers appear in the org P&L as approved expenses after admin approval.

9.2 Bill Vault

Searchable archive of all financial documents. Filters: Document type, Tenant name/Admission ID, Hostel, Date range, Amount range, Status.

S3 storage structure:

```
{orgId}/hostels/{hostelId}/tenants/{tenantId}/documents/{type}/{year}/{month}/{documentId}.pdf
```

10. Module 05: Expense Management

10.1 Expense Entities

Named, recurring expense obligations. Created exclusively by Main Admin.

Entity Definition: Name, description, applicable hostels (multi-select), authorized wardens (multi-select), expected frequency, approximate expected amount.

Example: 5 building rent entities — one per hostel, each scoped to its respective warden. NHP warden can only log "Building Rent — NHP."

10.2 Expense Tags

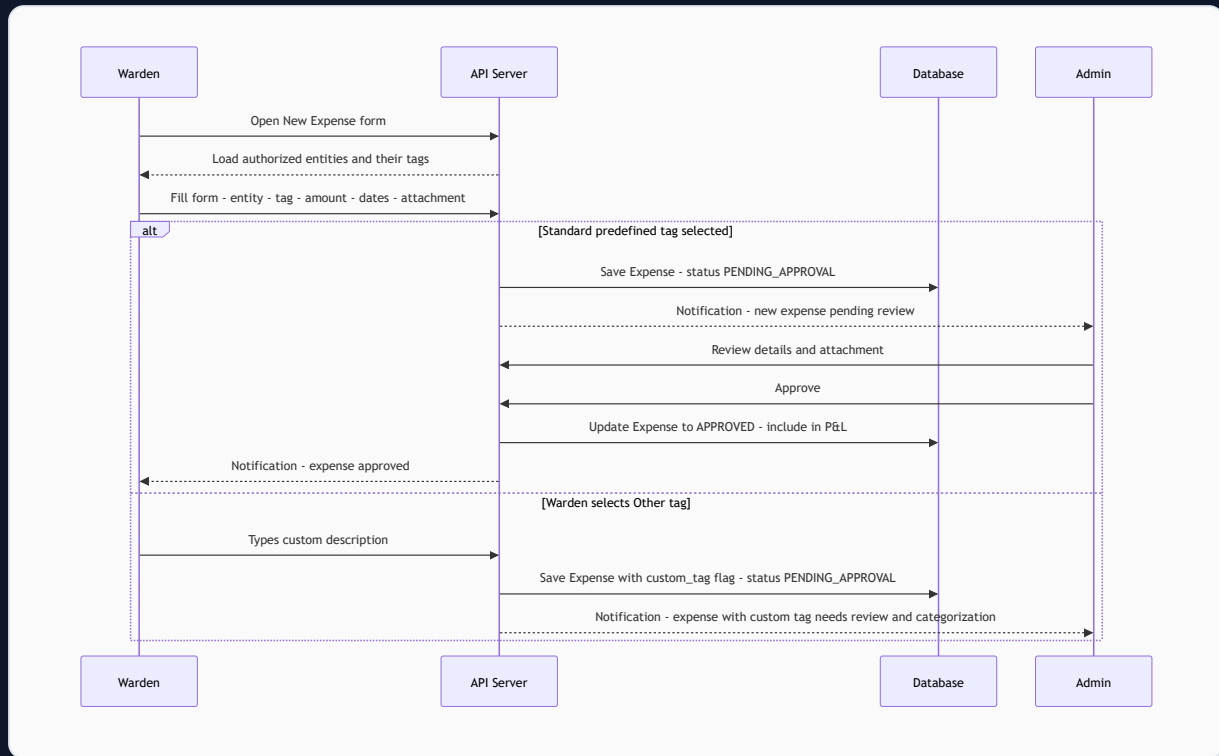
Sub-category labels created by Main Admin. Wardens must select from predefined tags — they cannot create new tags.

Examples:

- Entity: Electricity → Tags: "Meter 1 Ground Floor", "Meter 2 First Floor", "Common Area Meter"
- Entity: Water → Tags: "BWSSB Main Supply", "Borewell Maintenance"
- Entity: Others → Tags: "Hardware Purchase", "Plumbing Supplies", "Grocery Run"

If no tag fits: warden selects "Other" and types a description. Expense goes to admin for both financial approval and tag categorization (admin can create a new tag if this type recurs).

10.3 Expense Entry and Approval Flow



10.4 Expense Form Fields

FIELD	REQUIRED	NOTES
Expense Entity	Yes	Filtered by warden's access grants
Tag	Yes	Predefined list or "Other"
Custom Tag Description	If "Other"	Free text
Bill Date	Yes	Date bill was issued by vendor
Due Date	No	Payment deadline
Amount	Yes	
Attachment	Strongly recommended	PDF or phone camera photo
Notes	No	Additional context

11. Module 06: Food Management System

11.1 Per-Meal Pricing

Food is priced per individual meal (not per day as a package). Prices set per hostel by Main Admin:

- Breakfast price, Lunch price, Dinner price
- Add-on items: individually priced with name, availability window, and max quantity per tenant

11.2 Food Wallet — Prepaid Balance

Tenants maintain a prepaid Food Wallet. Top-up follows the standard payment flow (UPI/Cash + warden verification). When a meal is booked, the meal price is deducted. When balance falls below a configurable threshold, a low-balance notification fires.

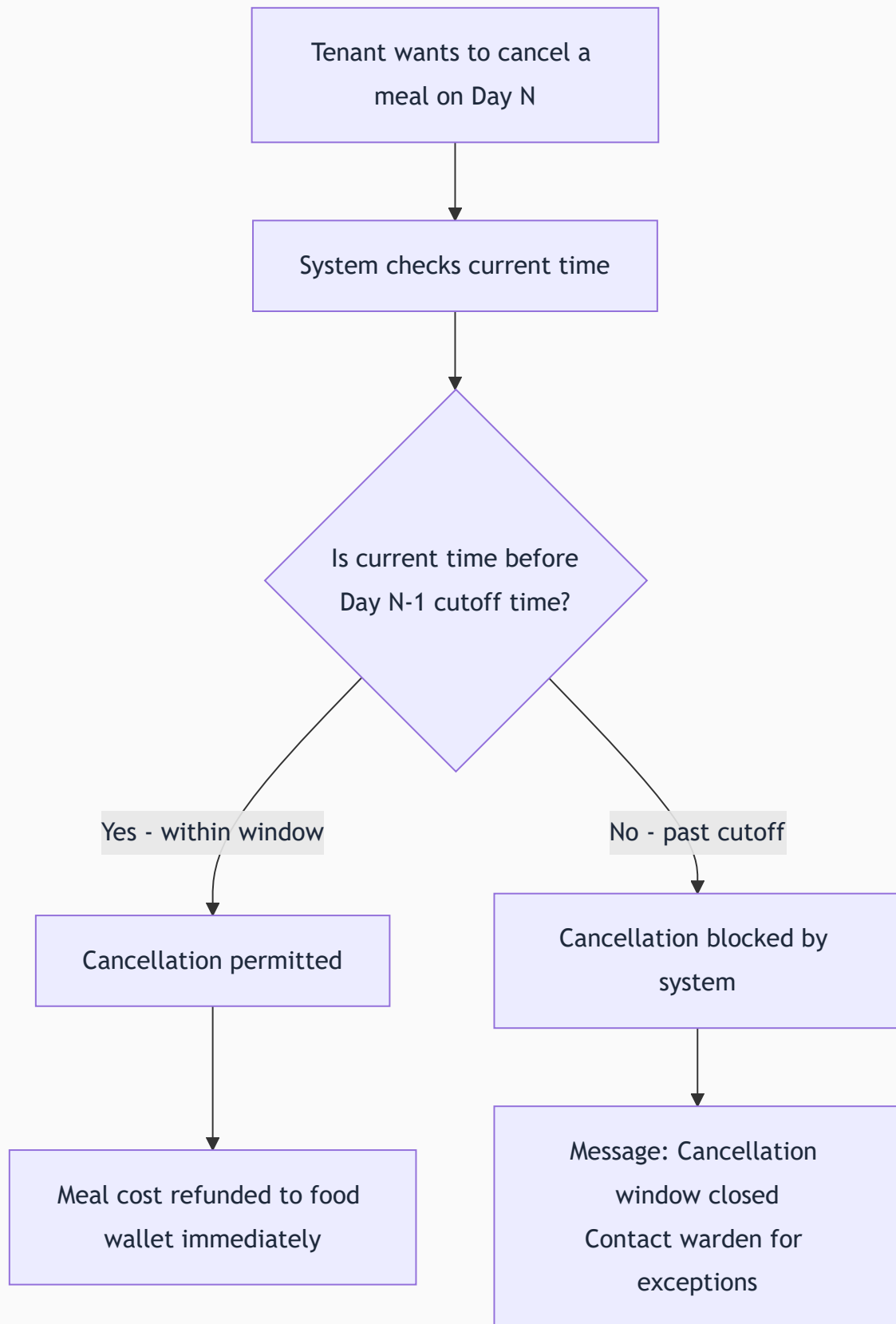
Wallet Transaction Log (per tenant):

- +Rs.2000 | Top-up | Verified by Warden Priya
- -Rs.80 | Breakfast booked Mon 21st July
- +Rs.80 | Breakfast cancelled Mon 21st July

11.3 Booking Rules

RULE	VALUE	CONFIGURABLE
Minimum booking period	7 days	No — fixed
Maximum booking period	28 days or remaining stay	No
Single-day ordering allowed	Yes — within cutoff window	No
Default cutoff time	10:00 PM previous night	Yes — admin per hostel

11.4 Cancellation Logic



The same cutoff window applies for both cancellations and single-day additions.

11.5 Admin Weekly Menu Management

Main Admin sets the meal menu for each day of the week: Breakfast item, Lunch item, Dinner item. Weekly menu repeats unless manually updated. Individual days can be overridden without changing the weekly template.

11.6 Tenant Food Calendar View

- Week/Month toggle
- Each day card: B / L / D indicators (orange = booked, gray = not booked)
- Tapping any day: bottom sheet showing menu items, prices, booking/cancellation actions (disabled if past cutoff)
- Food wallet balance displayed prominently at top

11.7 Food Procurement Report

Daily report (generated for tomorrow):

```
[Hostel Name] - Food Procurement Report | Date: [Tomorrow]
```

```
BREAKFAST:    44 portions
LUNCH:        40 portions (4 cancellations received)
DINNER:       44 portions
```

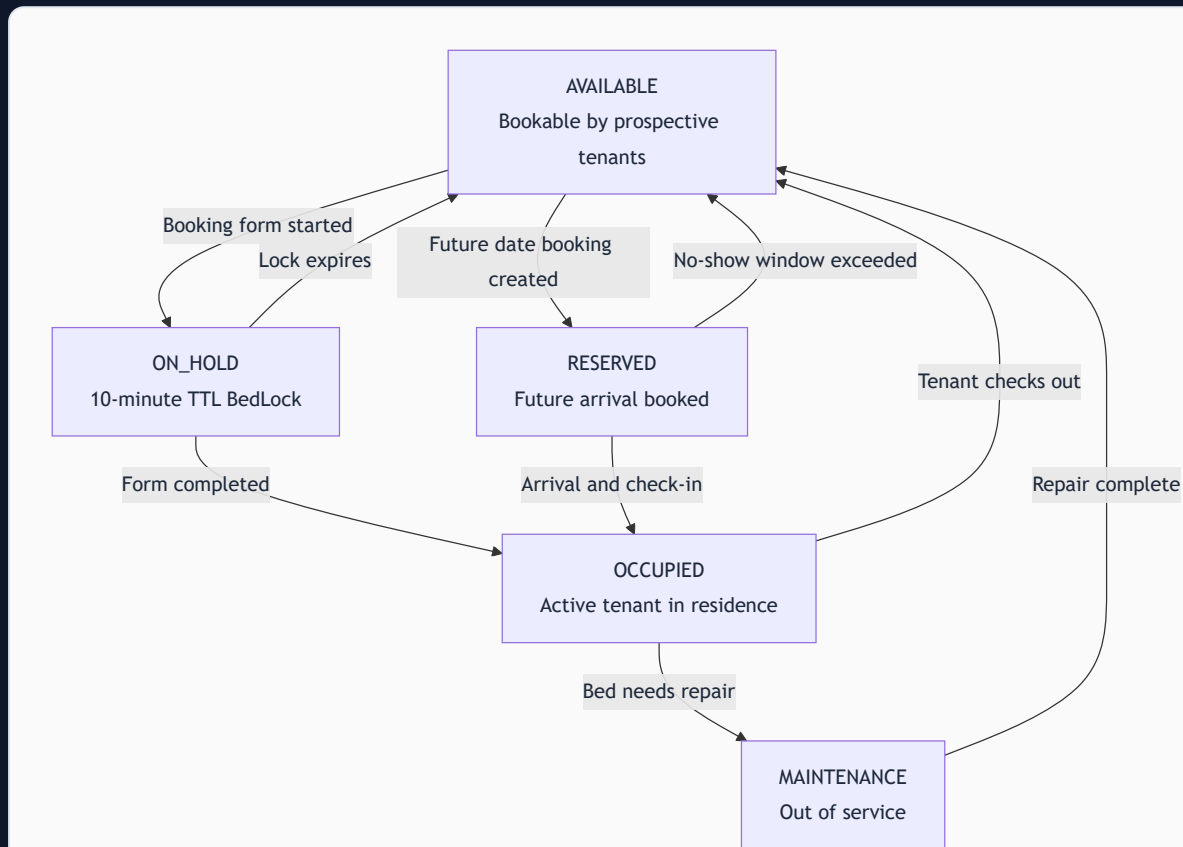
```
ADD-ONS:
  Protein Shake: 7 units
  Evening Snack: 12 units
```

```
ALERTS: 3 tenants wallet balance below Rs.300
```

Weekly report: Aggregated 7-day view. Access: Warden (own hostel), Main Admin (all hostels).

12. Module 07: Smart Bed Operations

12.1 Bed Status Model



12.2 Room QR Code Generation

Admin generates per room from Room Management panel. QR encodes a signed URL with a JWT scoped to that room. Token is permanent (room identity does not change). Downloads as high-resolution PNG for printing.

12.3 Bed Transfer

Move one tenant from their current bed to a different bed within the same hostel.

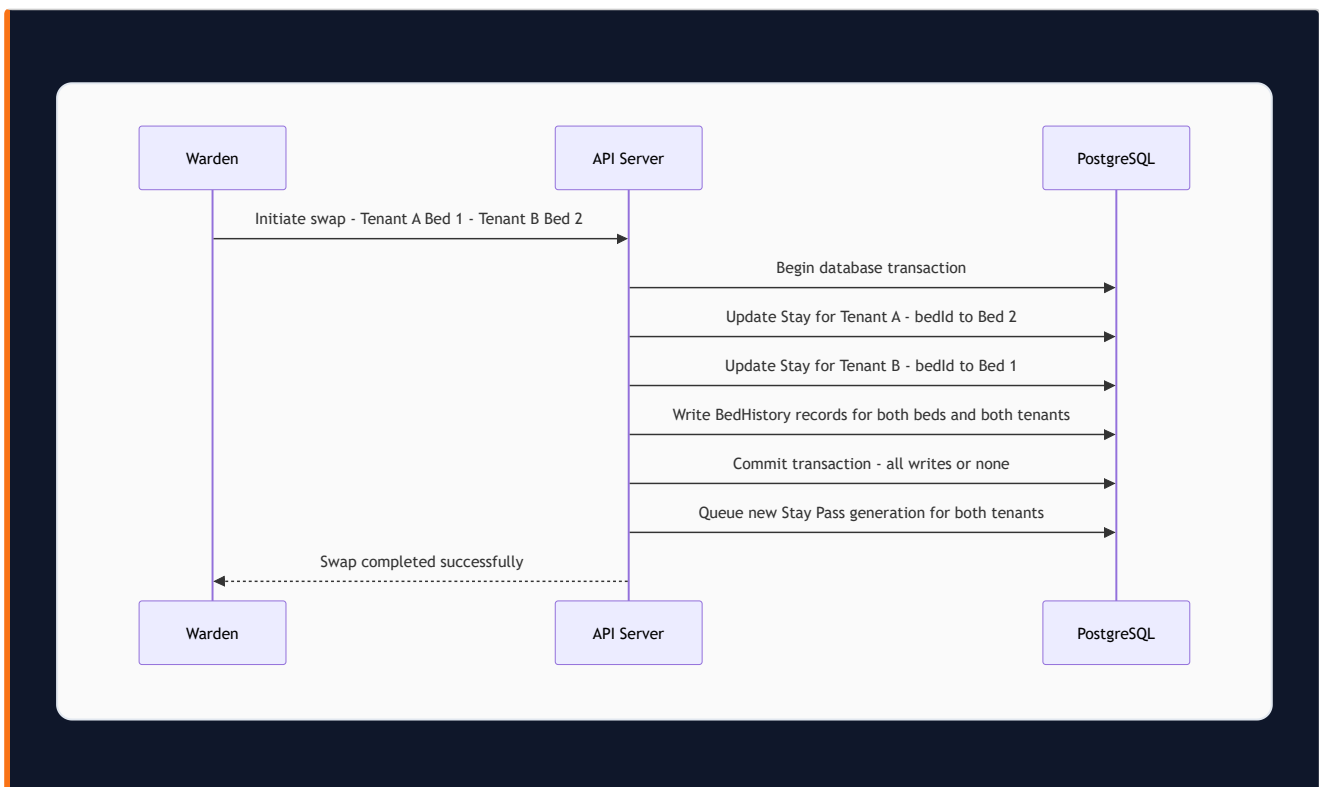
1. Warden selects tenant, clicks "Transfer Bed"

2. Selects destination bed (from AVAILABLE beds)
3. Selects effective transfer date and enters reason
4. System: source bed marked AVAILABLE, destination marked OCCUPIED, BedHistory updated for both with exact date ranges, new Stay Pass generated with updated bed details

Audit Guarantee: BedHistory records every occupancy period per bed permanently.

12.4 Bed Swap

Two existing tenants exchange beds atomically:



12.5 Bed Reservation (Future Date)

Admin/Warden creates a reservation: select bed, enter prospective tenant details, arrival date, expected stay duration. Bed status set to RESERVED. No-show release: configurable window (default 3 days after arrival date) after which bed auto-returns to AVAILABLE.

12.6 Waiting List

When a prospective tenant's QR scan shows all beds occupied:

- System shows "No beds available — Join Waiting List"
- Tenant enters Name, Phone, Email
- Record added to WaitingList (FIFO)

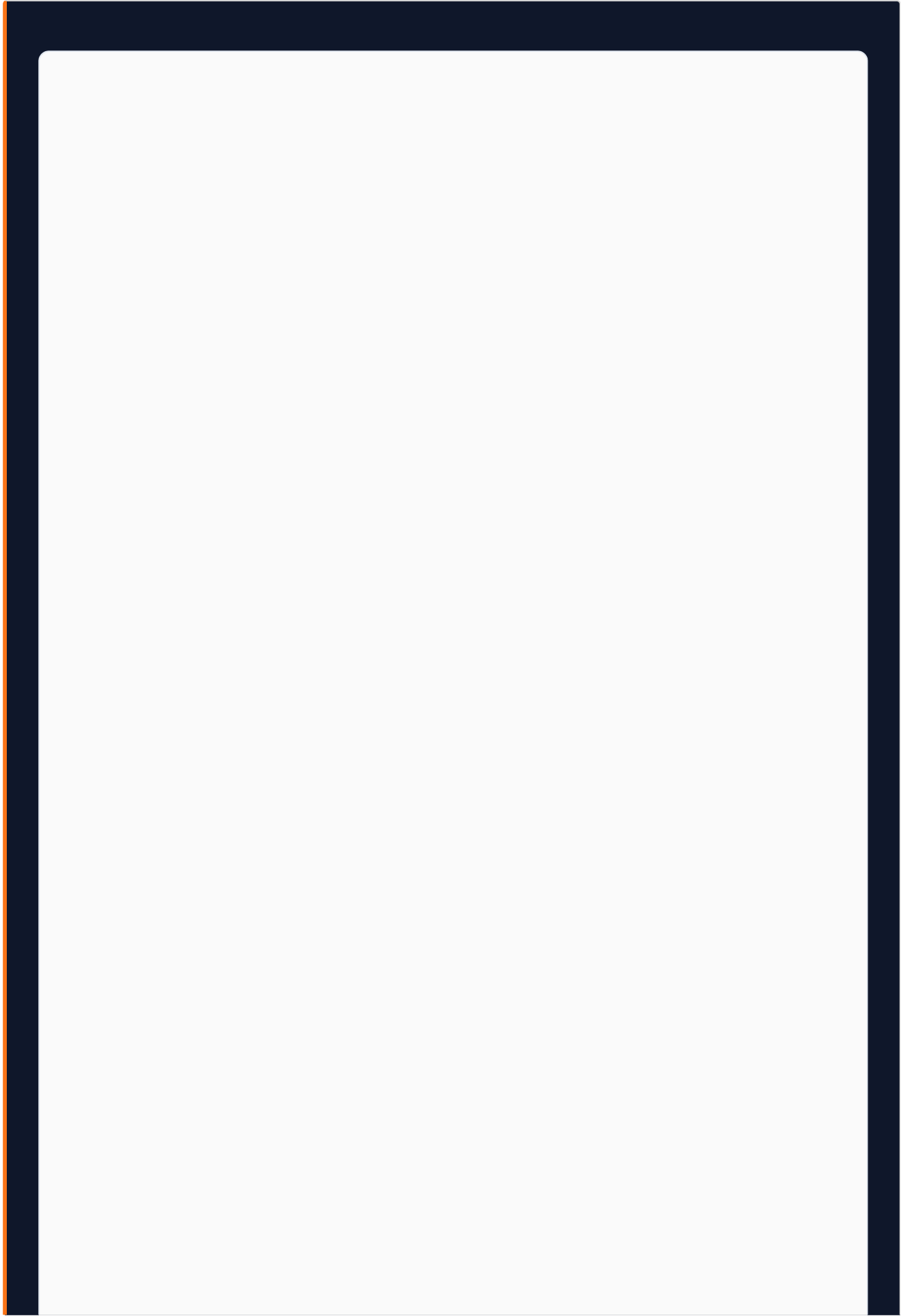
- When any bed becomes AVAILABLE: first person on list is notified, has a configurable response window before next person is notified
-

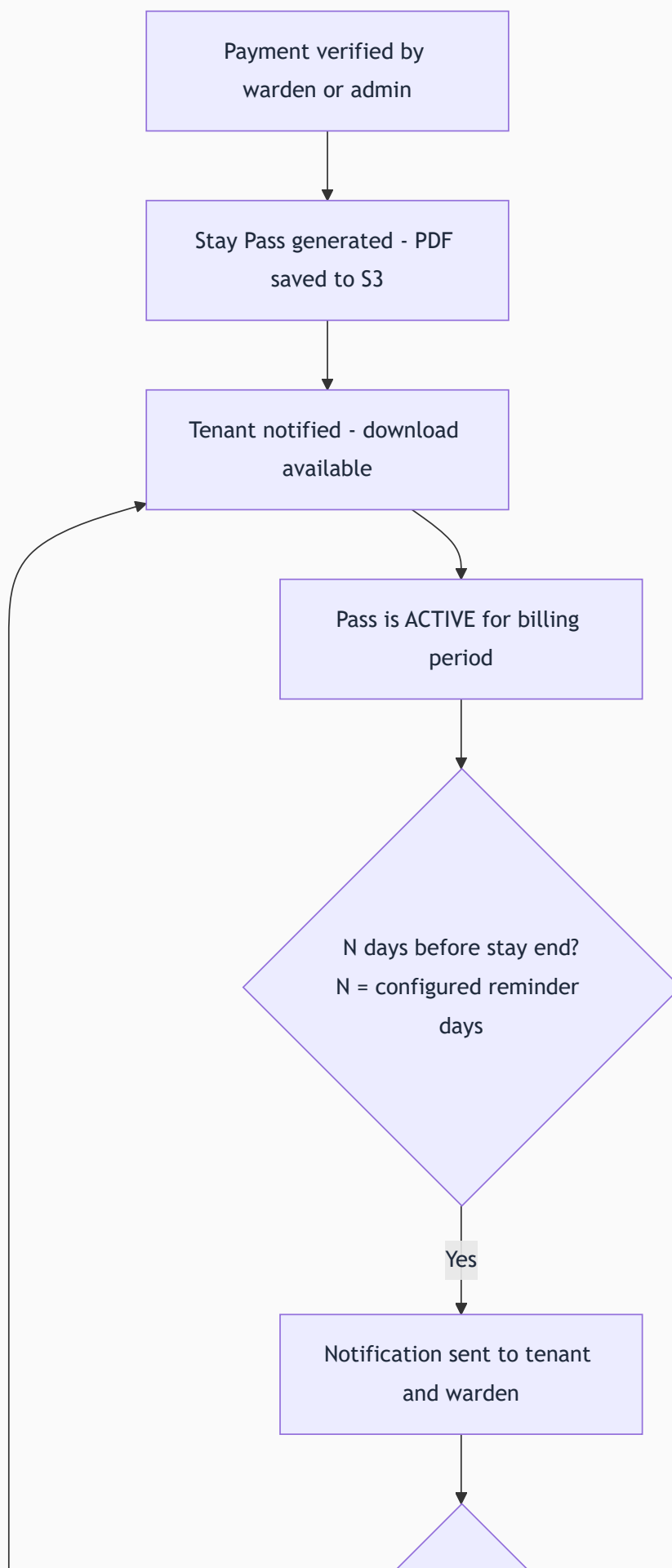
13. Module 08: Stay Pass System

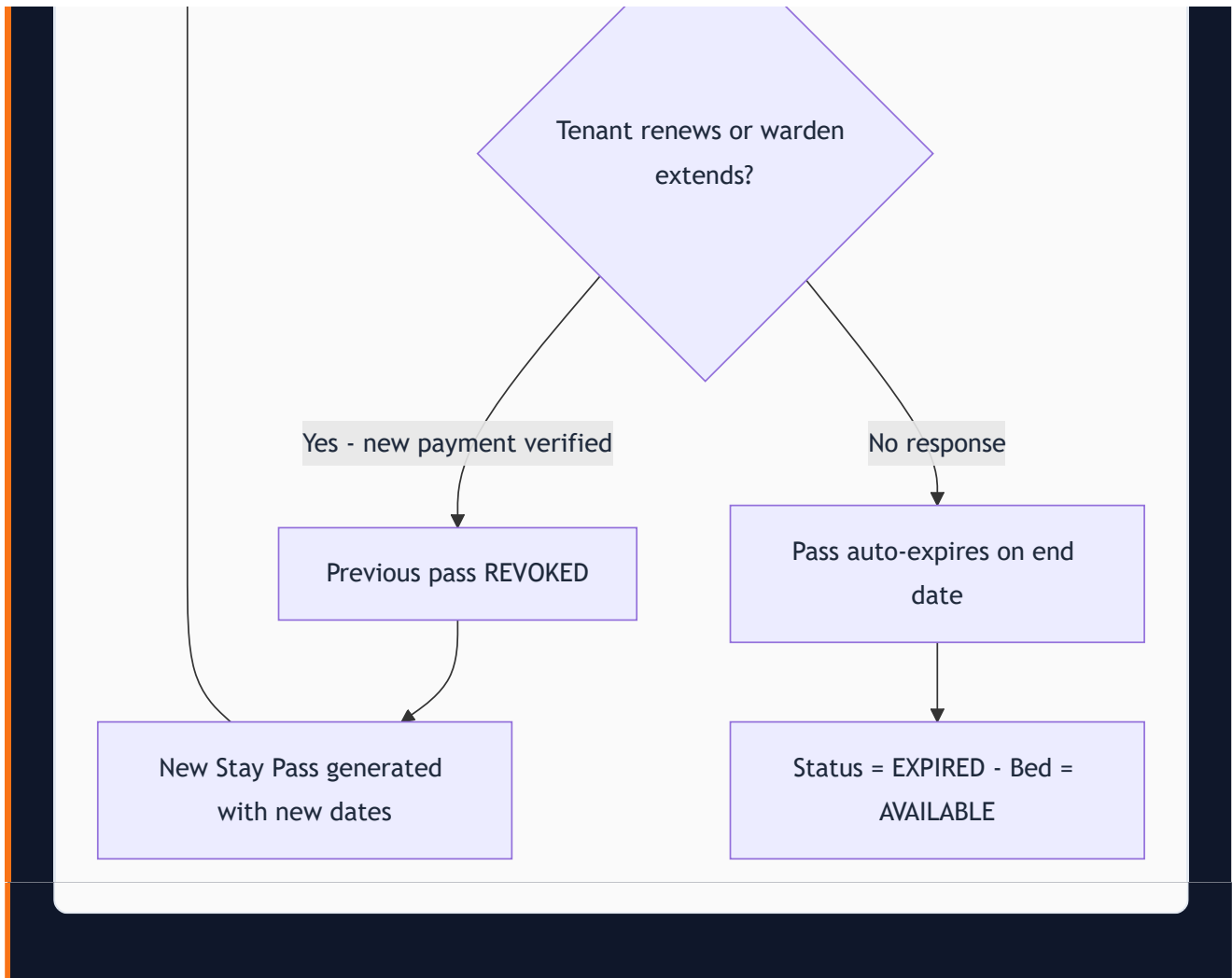
13.1 Stay Pass PDF Content

ELEMENT	DETAILS
Platform Logo	Top header with organization logo
Hostel Name and Address	Full address
Tenant Full Name	As submitted at onboarding
Tenant Photo	From onboarding upload
Admission ID	Auto-generated sequential unique ID
Room and Bed Number	e.g., Block G - Floor 1 - Room 01 - Bed A
Stay Period	Valid From date and Valid To date
KYC Verified	Document type shown (Aadhaar / PAN / License) — number NOT printed
Issue Date	
Issued By	Warden or Admin name
Verification QR Code	Links to real-time pass validity check page
Pass Status	VALID stamp (green)

13.2 Stay Pass Lifecycle





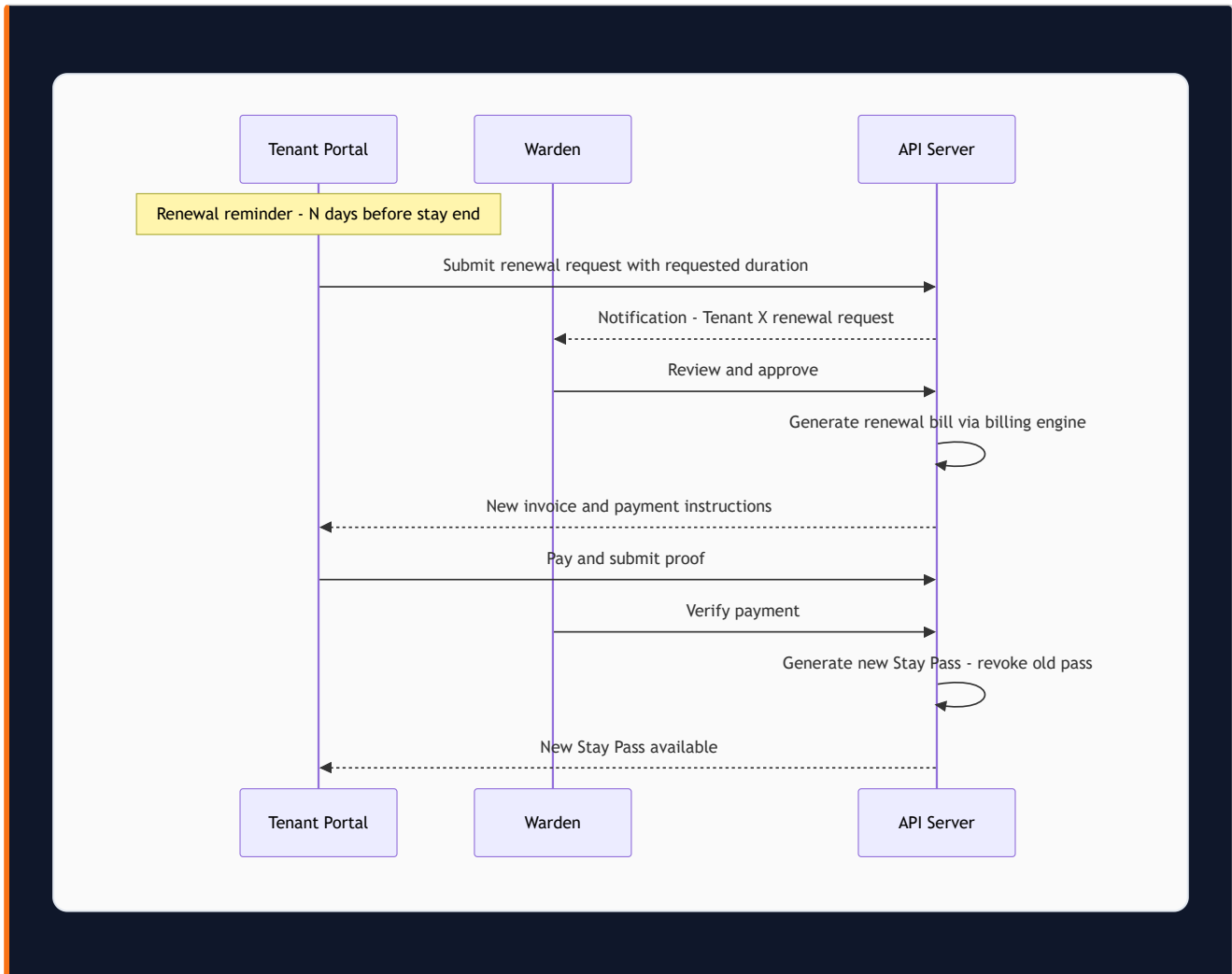


13.3 Pass Validity Check

QR on the pass links to a public verification URL showing: Tenant name, Hostel, Bed number, VALID (green) or EXPIRED (red), Valid until date. Scannable by any phone camera — no app required.

14. Module 09: Lease Renewal Workflow

14.1 Renewal Flow



14.2 Overstay Detection

If a tenant's stay end date passes without renewal or extension:

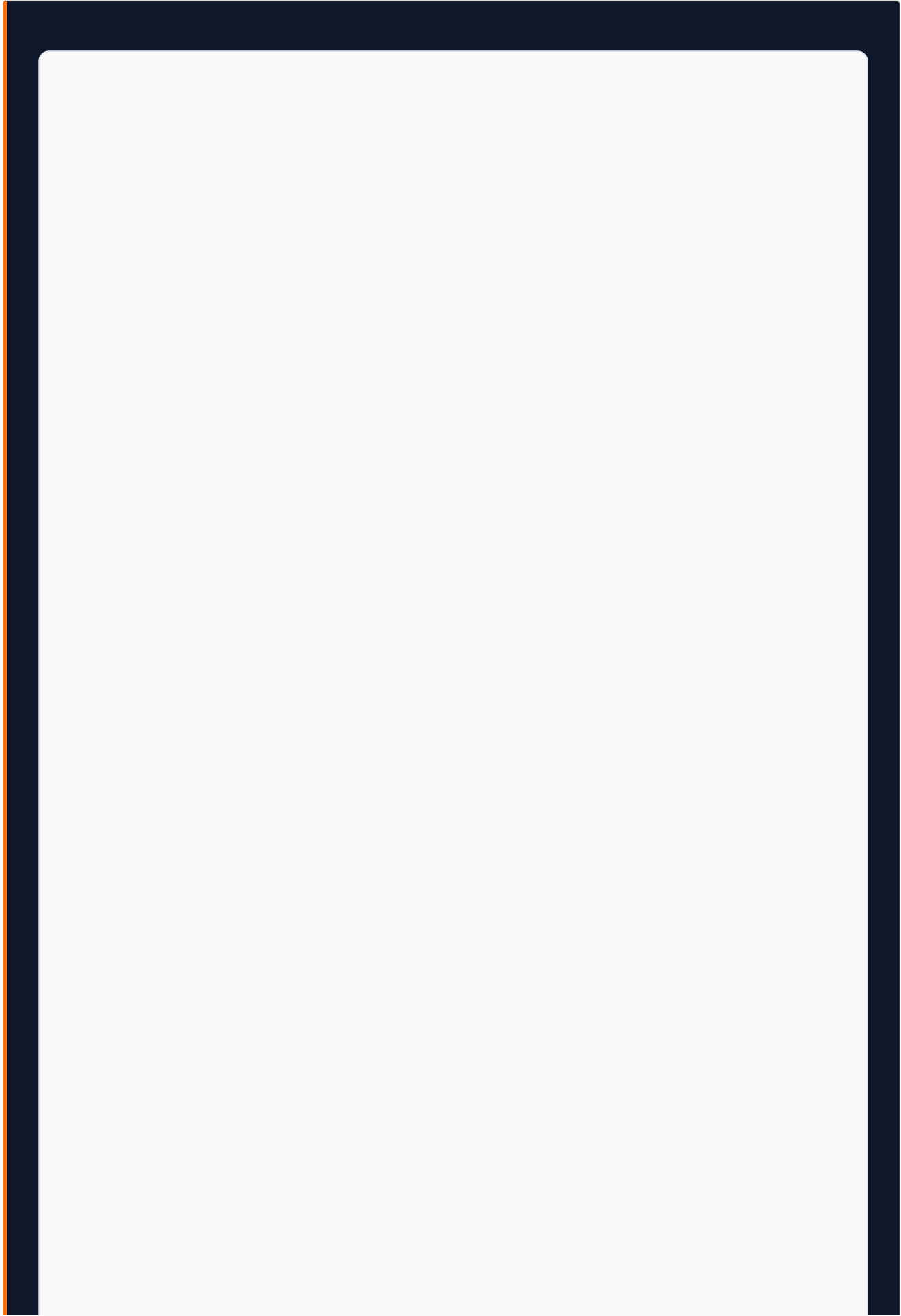
- Warden receives immediate notification: "Tenant X stay expired today — no renewal received"
- Tenant portal shows "Stay Expired" banner
- Stay Pass status automatically changes to EXPIRED
- Bed marked AVAILABLE in the system
- Configurable grace period (default 3 days) allows manual extension without full re-onboarding

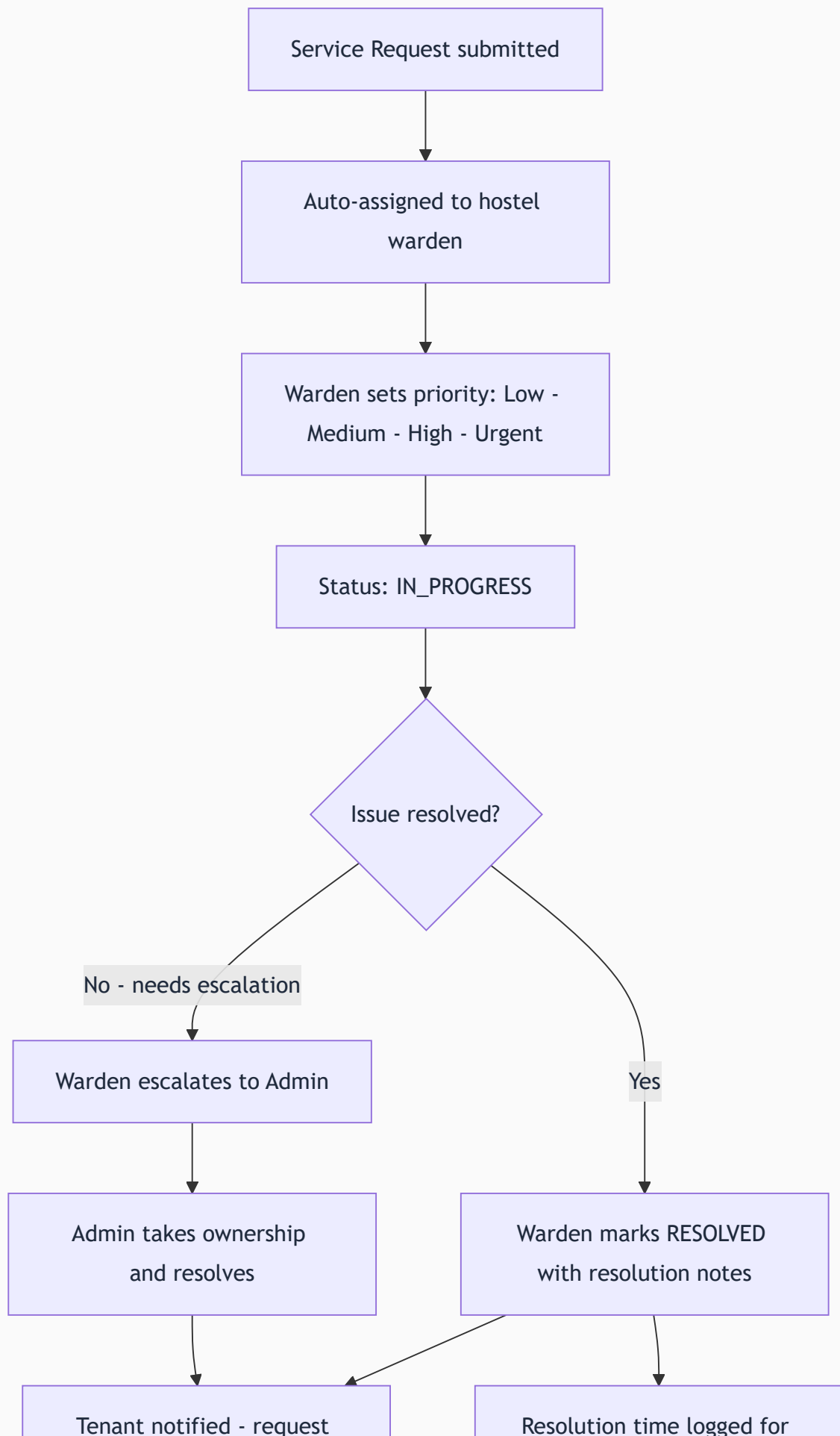
15. Module 10: Service Requests

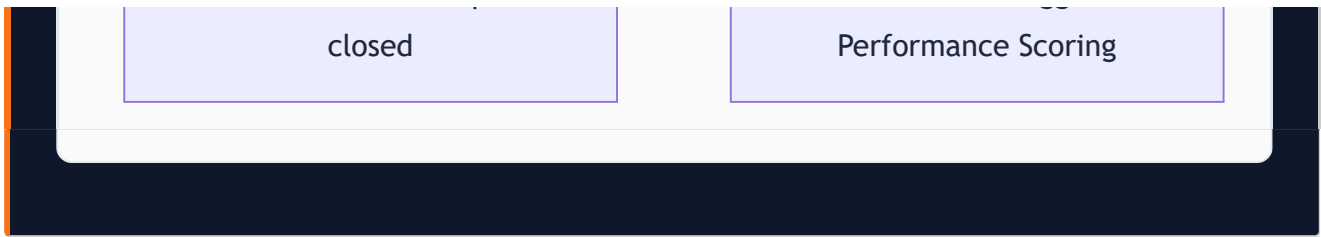
15.1 Categories

CATEGORY	WHO CREATES	ROUTING	NOTES
Maintenance Request	Tenant	Assigned to hostel warden	AC broken, tap leaking, etc.
Complaint	Tenant	Assigned to hostel warden	Service quality, food, noise, etc.
Incident Report	Warden	Escalated to Admin	Safety, property damage, major events

15.2 Service Request Lifecycle



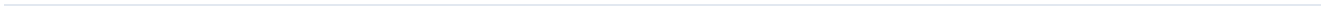




15.3 Fields

Category, Title (max 100 chars), Description, Photo Attachment (optional), Priority (set by warden/admin), Status (OPEN / IN_PROGRESS / ESCALATED / RESOLVED / CLOSED), Resolution Notes, Resolution Time (auto-calculated).

SLA: Unresolved requests older than 7 days trigger automatic escalation notification to Admin.



16. Module 11: Visitor & Guest Log

16.1 Log Entry Fields

FIELD	REQUIRED	NOTES
Visitor Full Name	Yes	
Visitor Phone Number	Yes	
Tenant Being Visited	Yes	Dropdown from active tenants
Relationship	No	Family / Friend / Professional / Other
Purpose of Visit	No	
ID Proof Type	Recommended	Aadhaar / License / Passport
Entry Time	Yes	Auto-filled from system clock
Expected Exit Time	No	
Actual Exit Time	Yes	Filled when visitor leaves

16.2 Access and Compliance

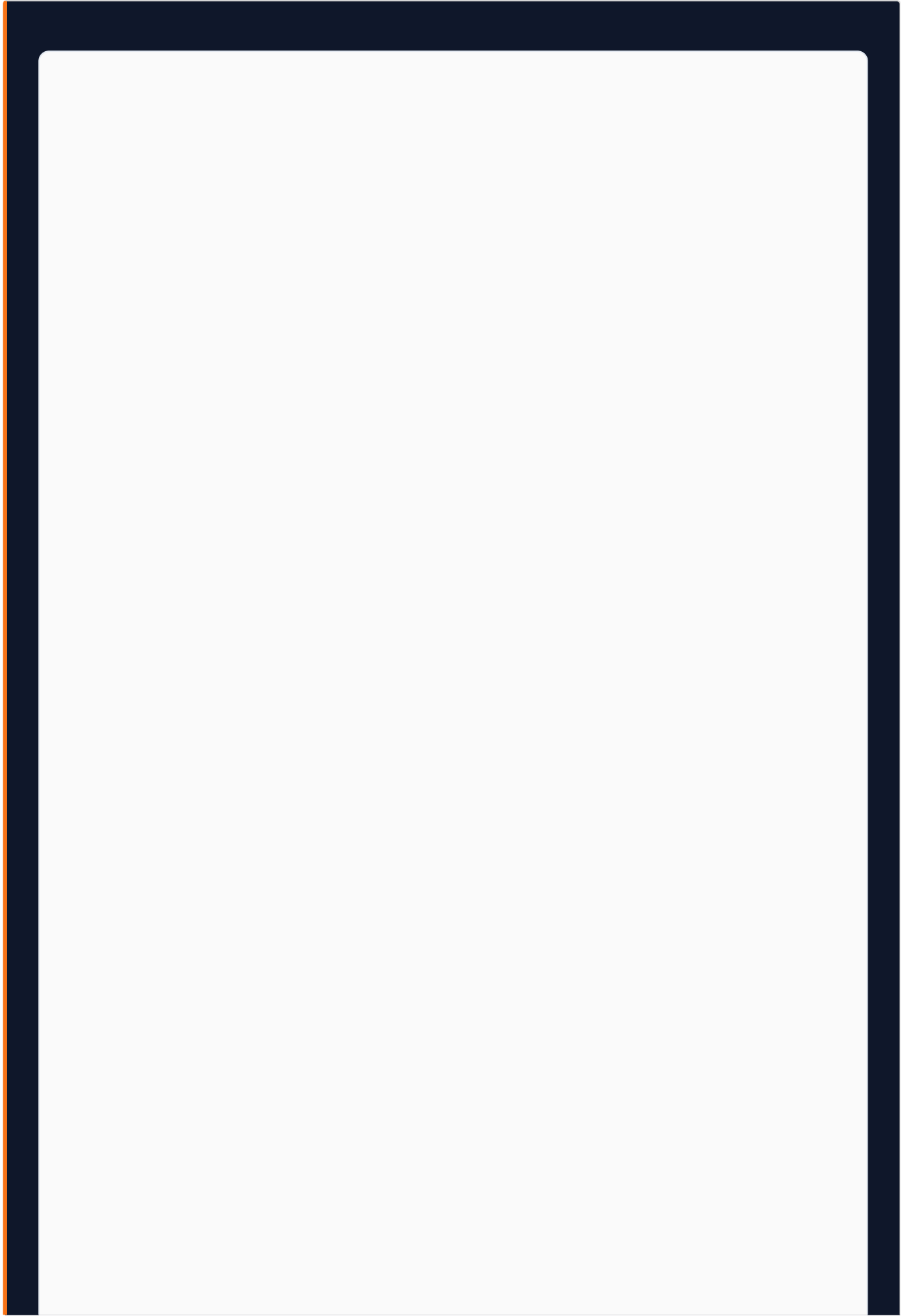
- **Warden:** Create entries and view all logs for their hostel
- **Main Admin:** View all logs across all hostels
- **Filters:** Date range, tenant visited, hostel
- **Export:** PDF export in compliance-ready format
- **Retention:** 12 months active, then archived (not deleted)

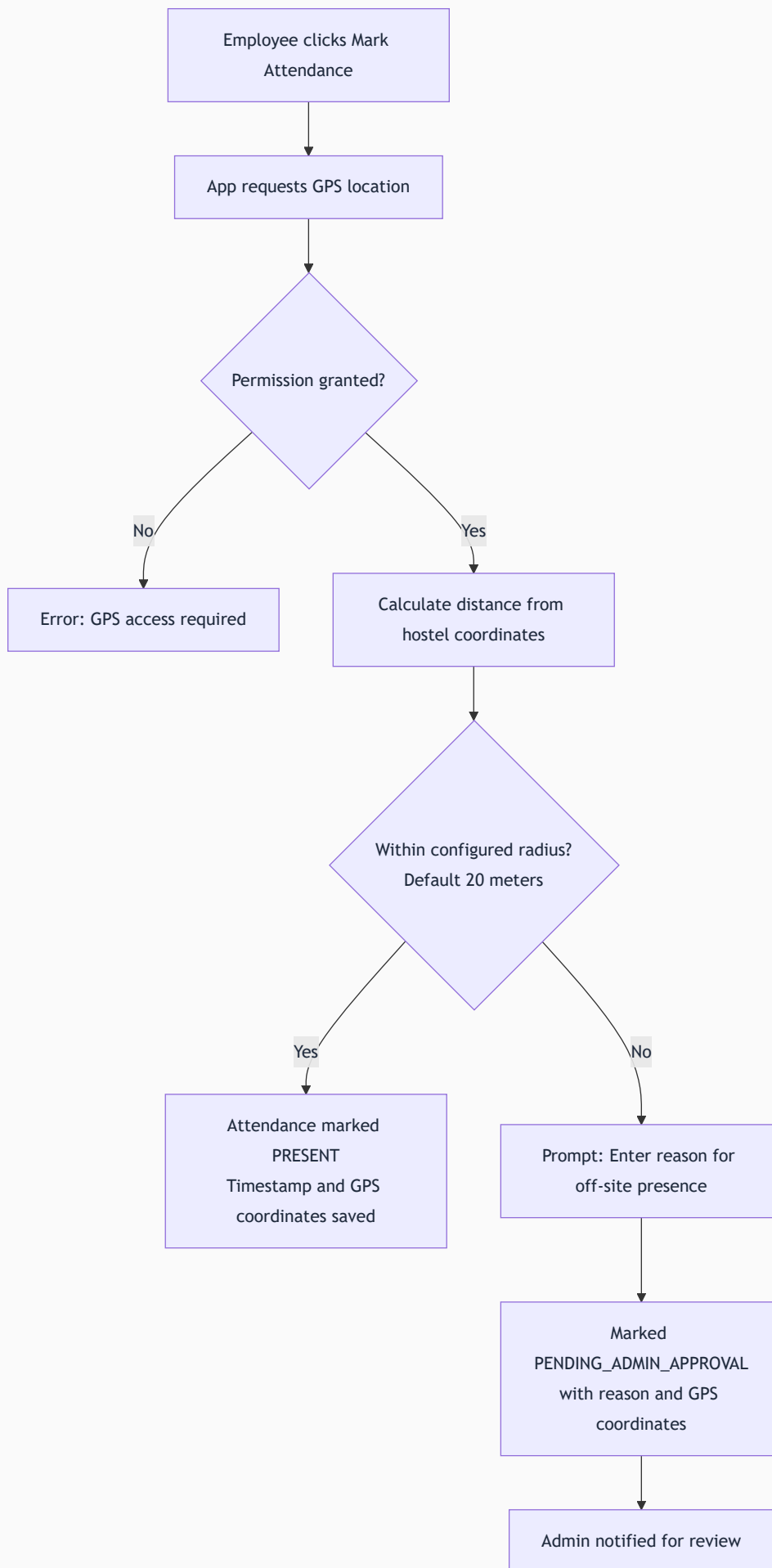
17. Module 12: Employee & Attendance Management

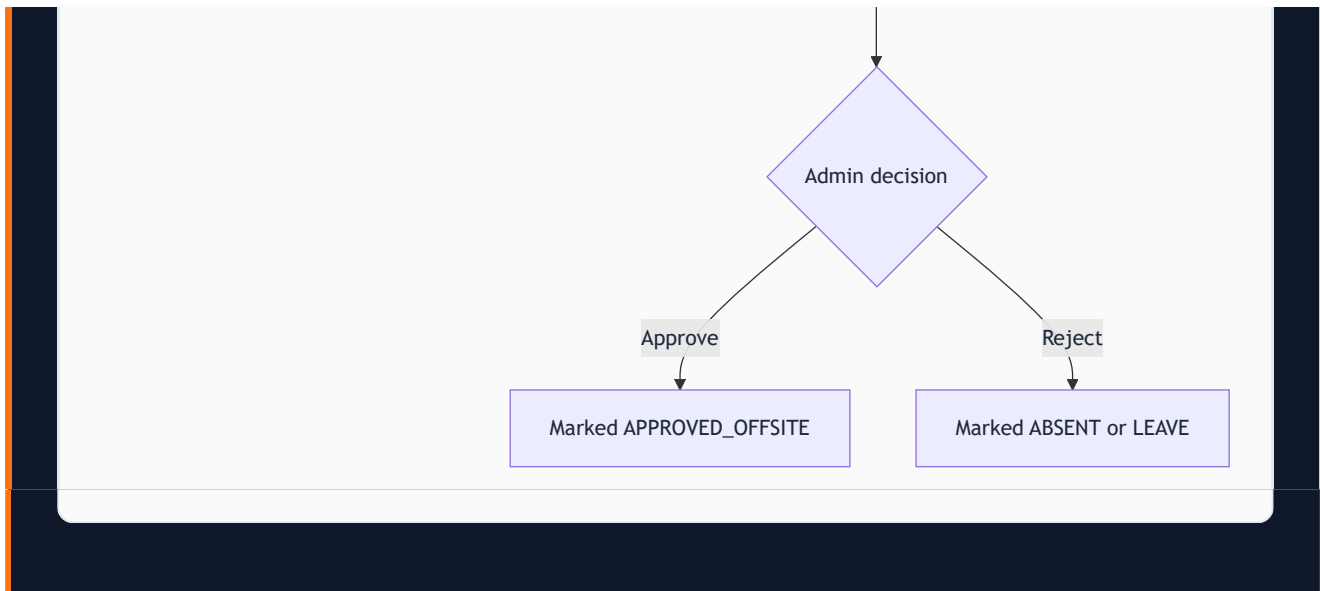
17.1 Employee Onboarding

Added by Main Admin or hostel Warden. Fields: Full Name, Phone (login ID), Email (optional), Designation, Assigned Hostel, Work Start Time, Work End Time, Date Joined.

17.2 Geolocation Attendance Flow







17.3 Employee Portal Scope

FEATURE	ACCESS
Mark attendance	Yes
Mark checkout	Yes
View own attendance history (calendar)	Yes
View own working hours log	Yes
Submit leave request	Yes
View own leave request status	Yes
View assigned tasks	Yes
Update task status (In Progress / Completed)	Yes
Upload task completion photo	Yes
Access any financial or tenant data	No

17.4 Leave Management

Employees submit leave requests with: date(s), leave type (Casual / Sick / Emergency / Other), reason, optional supporting document.

Only the Main Admin can approve or reject leave requests. Wardens have no access to this module.

After admin decision: Approved = day marked as LEAVE on attendance calendar. Rejected = admin adds rejection reason.

17.5 Task Assignment

Admin or hostel Warden assigns tasks to employees: employee selection, title, description, category (Housekeeping/Maintenance/Delivery/Other), due date, priority, optional linked room. Employees see and update tasks in their portal.

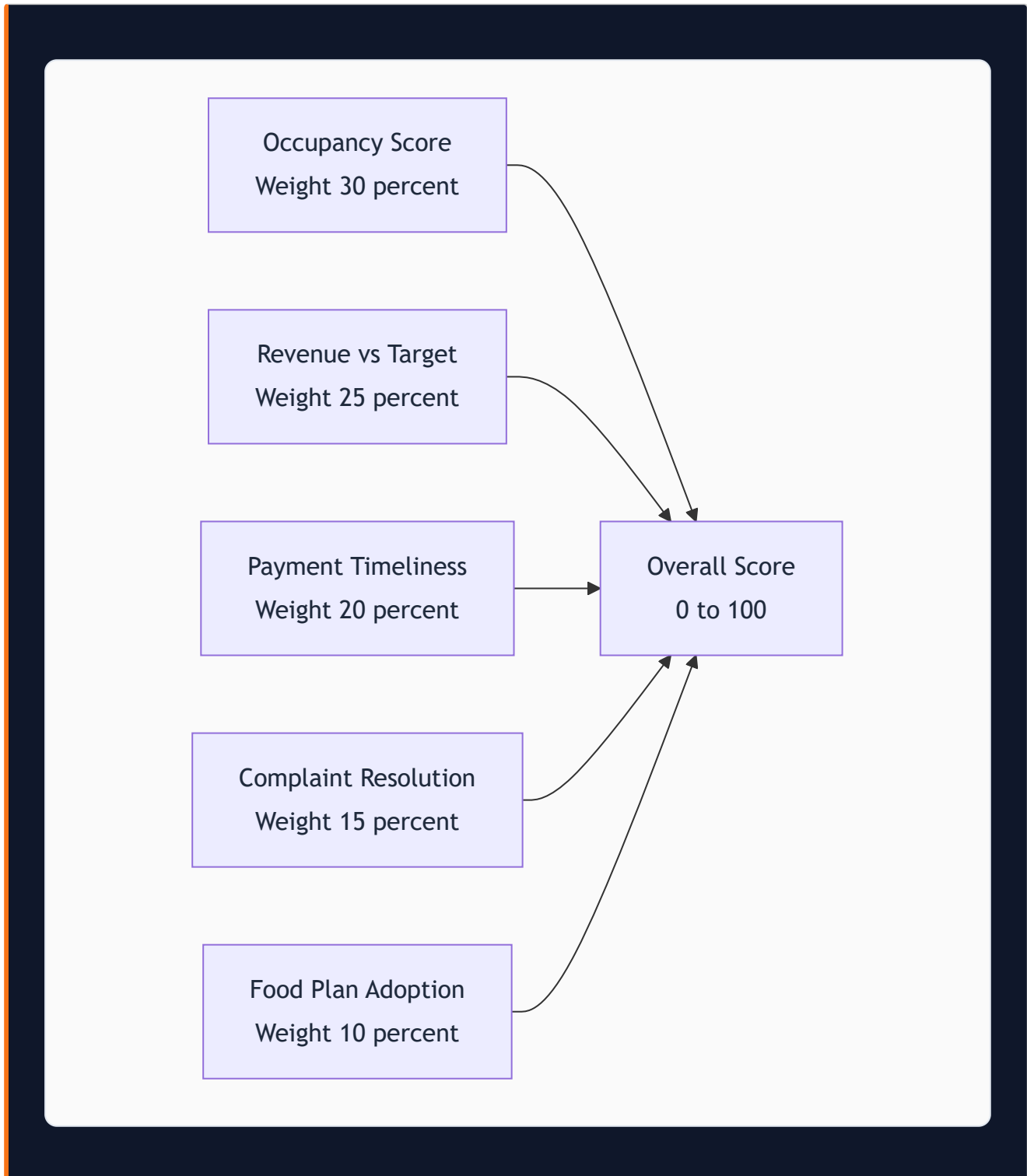
17.6 Attendance Reports (Admin Only)

Monthly attendance summary per employee (Present/Absent/Leave/Half-Day/Pending counts), working hours per employee, hostel-level attendance overview.

18. Module 13: Performance Scoring Engine

18.1 Scoring Architecture

Each hostel receives a Performance Score (0 to 100) computed from five weighted components:



18.2 Component Calculations

COMPONENT	FORMULA
Occupancy Score	(Occupied Beds / Total Beds) x 100
Revenue vs Target	(Actual Verified Revenue / Target Revenue at full occupancy) x 100, capped at 100
Payment Timeliness	(Payments made within grace period / Total payments due) x 100
Complaint Resolution	(Issues resolved within 48 hours / Total issues) x 100. Unresolved issues older than 7 days = 0 for this component
Food Plan Adoption	(Tenants with at least one meal booked this month / Total active tenants) x 100

18.3 Transparent Score Display

NextHome Paradise

Performance Score: 74/100 [Amber]

Occupancy

82/100 [Good]

22 of 27 beds occupied

Revenue vs Target

78/100 [Good]

Rs.154,000 of Rs.189,000 target

Payment Timeliness

65/100 [Average]

4 tenants paid outside grace period

Issue Resolution

55/100 [Low]

Avg resolution: 3.2 days

Food Adoption

88/100 [Excellent]

19 of 22 tenants on food plan

Monthly Trend: Score decreased 6 points from last month

Main drag: Issue Resolution time increased from 1.8 to 3.2 days

Suggestion: 2 service requests are 5+ days old and unresolved

18.4 Score Color Bands

RANGE	COLOR	MEANING
80 to 100	Green	Excellent
60 to 79	Amber	Needs attention in specific areas
Below 60	Red	Critical — immediate focus required

19. Module 14: Comparisons & Reporting

19.1 View Modes

Consolidated View: Aggregates all hostels into one organization-level view.

Compare Mode: Admin selects 2 or more hostels for side-by-side comparison.

Available Metrics: Monthly Revenue, Net Profit, Occupancy Rate, Payment Timeliness, Food Adoption, Overall Performance Score, Complaint Volume, Average Resolution Time.

Date Range: Last 7 days / Last 30 days / Last 3 months / Custom range.

19.2 Report Output

Downloadable as **PDF only** (V1). Report contains:

- Organization name, logo, generation timestamp
- Date range and hostels compared
- Side-by-side values and percentage differences for each metric
- Trend charts per metric
- Narrative summary: "NHP generated 23% more revenue than NHC in June 2026. However, NHC has a 12% higher food plan adoption rate."

Access: Main Admin only. Wardens cannot access this module.

20. Module 15: Announcements

20.1 Creation

Main Admin: Can broadcast to all hostels or select specific hostels. **Warden:** Can broadcast to tenants of their own hostel only.

Fields: Title, Body/Message, Type (General / Food / Maintenance / Urgent), Target hostel(s), Expiry date.

20.2 Delivery Channels (V1 — In-App Only)

- Banner at top of tenant portal (for active/unexpired announcements)
- In-app notification entry
- Dedicated "Announcements" section with full history

20.3 Examples

- **[General]** "Water supply off Sunday 20th July 8 AM to 12 PM — tank maintenance"
 - **[Food]** "No dinner service this Saturday — kitchen deep cleaning"
 - **[Maintenance]** "Elevator under maintenance — use staircase"
 - **[Urgent]** "Emergency: All tenants please evacuate the first floor immediately"
-

21. Module 16: Inventory & Asset Management

21.1 Asset Record Fields

FIELD	NOTES
Asset Name	e.g., "Daikin 1.5 Ton AC — Room G01"
Category	Furniture / Appliance / Electronics / Linen / Kitchen / Other
Hostel	Which hostel
Location	Floor, room, or common area
Purchase Date	
Purchase Cost	
Current Condition	Good / Fair / Poor / Out of Service
Serial Number or Asset Tag	Optional
Photo	Optional
Notes	

When an asset undergoes repair, the associated expense entry can be linked to it for lifecycle cost tracking.

Access: Main Admin (all hostels), Warden (own hostel only).

22. Module 17: Housekeeping Management

22.1 Schedule Creation

Warden or Admin creates housekeeping schedules: select room(s) or area, select frequency (Daily/Weekly/On-demand), assign to a specific employee, set preferred time window.

22.2 Photo Verification Protocol

Employee opens assigned task, clicks "Start Task" and uploads **Before photo** (via phone camera). Completes cleaning. Clicks "Complete Task" and uploads **After photo**. Both stored in S3, linked to the task record. Warden views before/after comparison for verification.

22.3 Reports

Weekly completion report: rooms cleaned, by whom, at what time, and any overdue or skipped tasks. Accessible by Warden (own hostel) and Main Admin (all hostels).

23. Module 18: Settings & Configuration

23.1 Organization Settings (Main Admin Only)

SETTING	DESCRIPTION
Organization Name and Logo	Displayed on all documents
UPI ID	Organization-wide UPI ID for all tenant payments
UPI QR Image	Upload existing or auto-generate from UPI ID
Default Currency	INR (default) or AED
Additional Currencies	Enable AED or others for display switch
Tax Groups	Create, edit, manage
Late Fee Policy	Enable/disable, grace period, fee type
Stay Expiry Notification Days	Days before expiry to send renewal reminders
No-Show Release Window	Days before auto-releasing an unchecked reservation

23.2 Hostel Settings

SETTING	SET BY	DESCRIPTION
Bill Normalization Toggle	Admin (any hostel) or Warden (own hostel)	Align billing to 1st of month
Food Cutoff Time	Admin	Next-day order/cancel deadline
Food Pricing	Admin	Breakfast, lunch, dinner prices
Geofence Radius	Admin	Employee attendance proximity (default 20m)
Low Wallet Alert Threshold	Admin	Minimum food wallet balance before alert
Hostel Coordinates	Admin	Latitude/longitude for geofence

23.3 Currency Display

Currency is **display-only in V1** (no actual multi-currency transactions). Admin sets default display currency. Exchange rate is entered manually by admin and updated manually when needed. All financial figures respect the active currency selection.

24. Module 19: Social Authentication

24.1 Overview

Social login is available **exclusively on the Tenant Portal**. Admin and Warden logins remain email/password only for security.

24.2 Supported Providers (V1)

PROVIDER	IMPLEMENTATION
Google	Supabase Auth Google OAuth 2.0
Facebook	Supabase Auth Facebook OAuth

24.3 Logic

Returning tenant — account exists: Social OAuth flow completes, Supabase matches email to existing account, session created.

New user — no account: OAuth completes, no matching account found, tenant redirected to QR booking entry point with name and email pre-filled. Cannot self-register outside the booking flow.

Password recovery: V1 has no email-based reset (no email service configured). Tenants contact warden directly. Warden or Admin resets via admin panel.

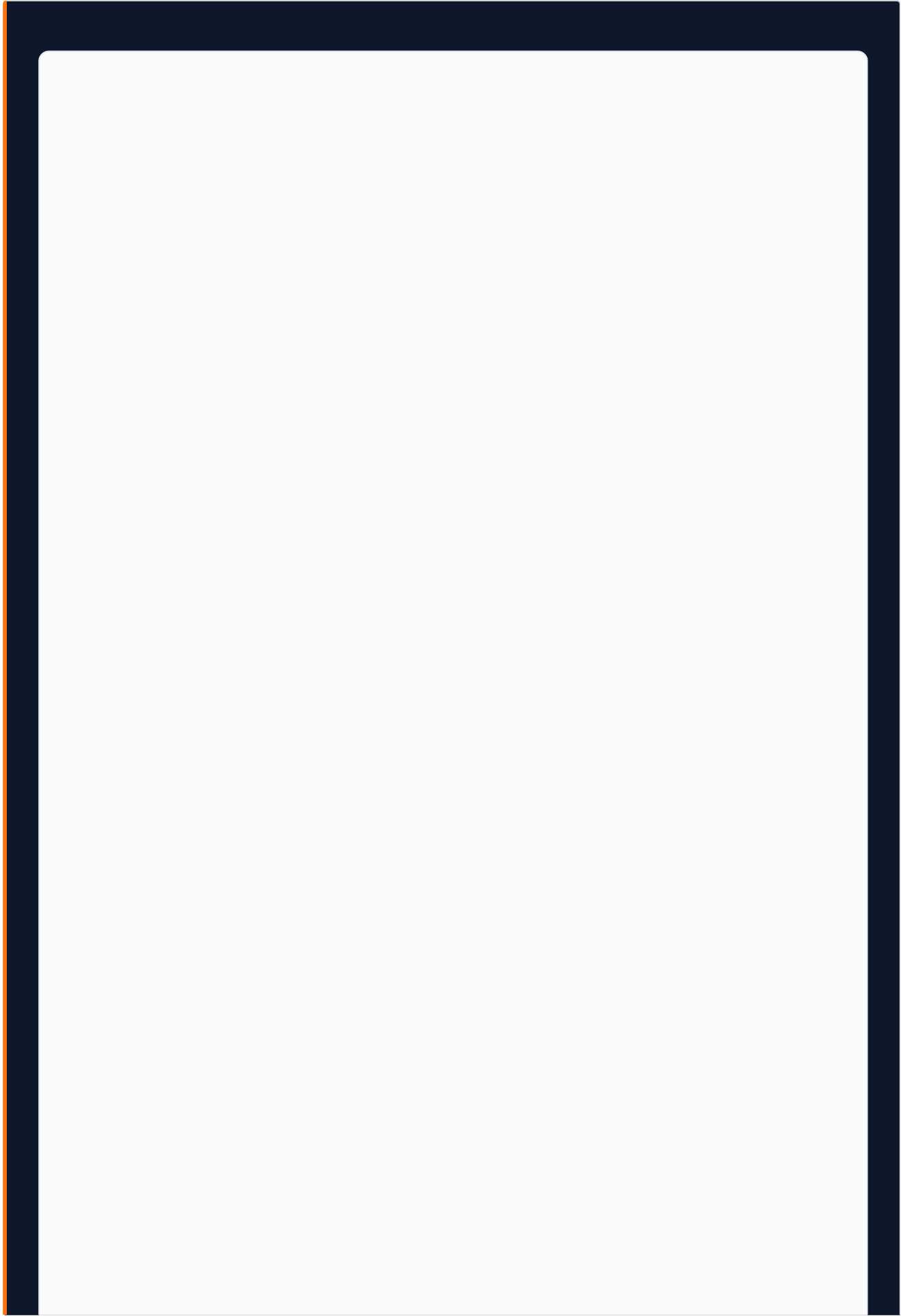
25. Module 20: DPDP Compliance & Privacy

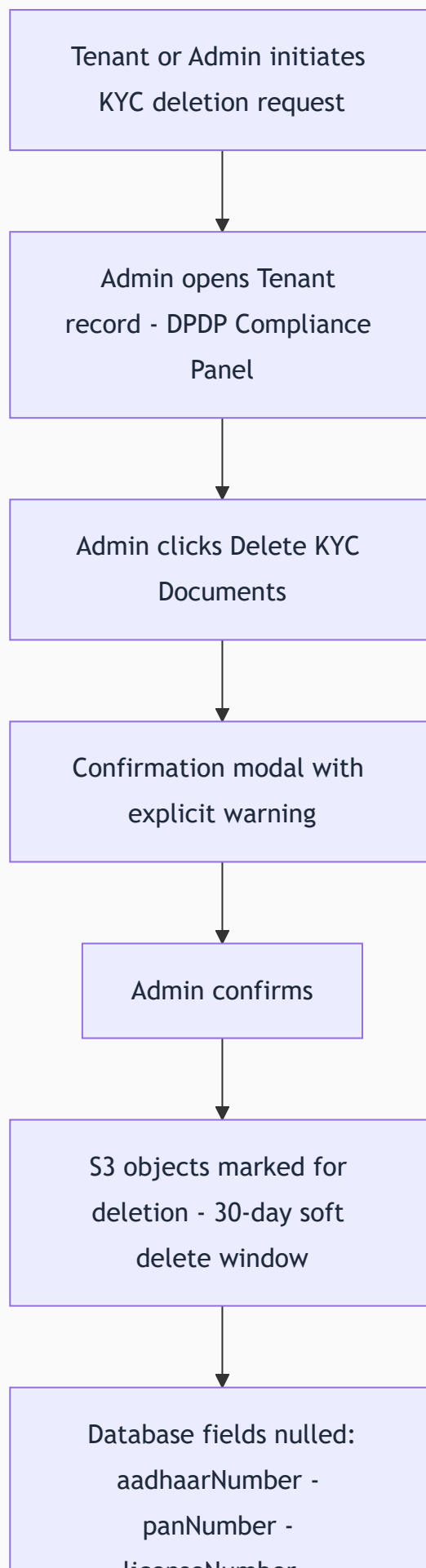
25.1 Consent Collection

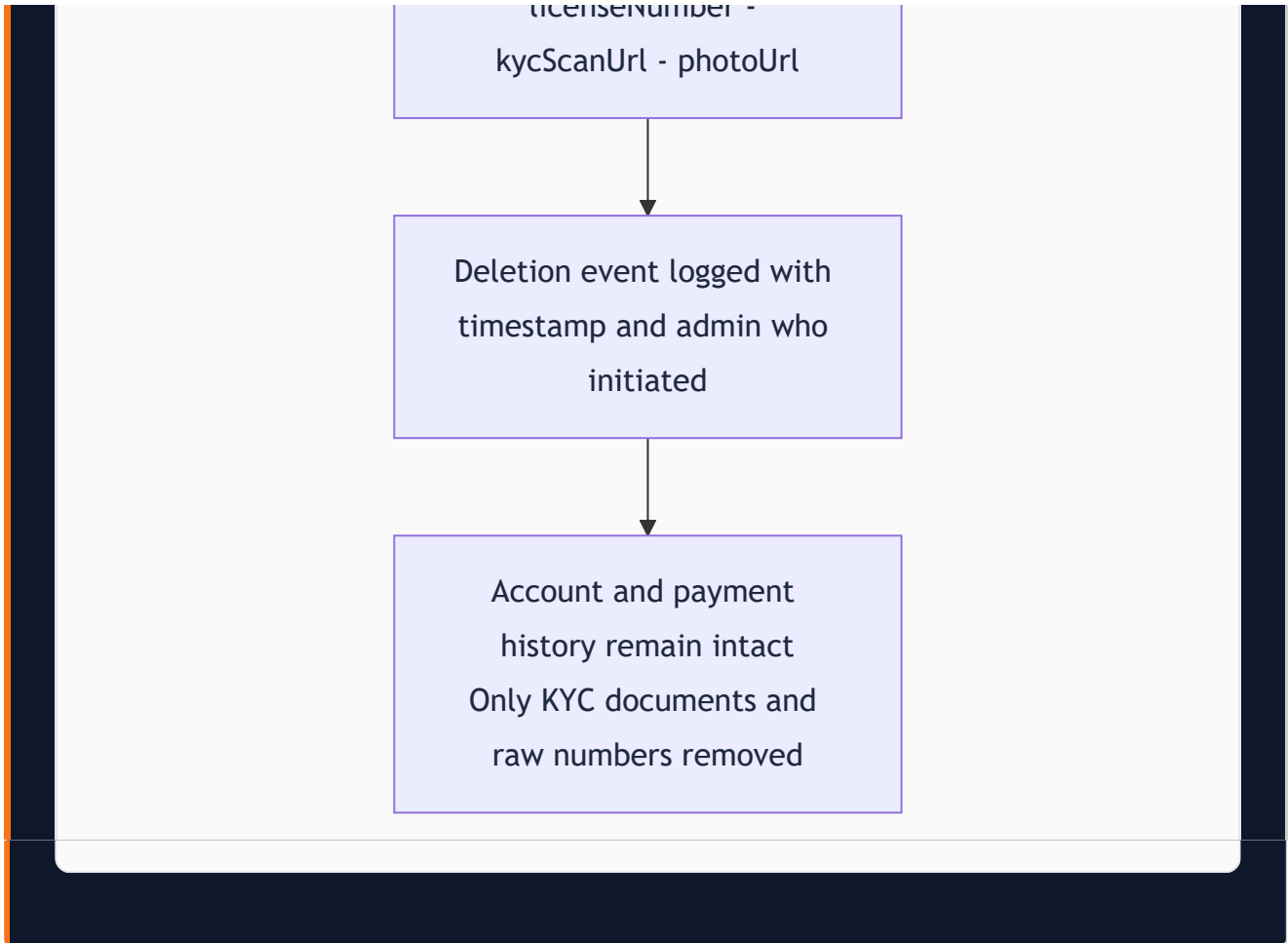
At the onboarding form, before any personal data fields appear, a clear consent section is shown listing exactly what data is collected and why. Tenant checks a mandatory checkbox. Submission is blocked if unchecked.

Consent record stored: `{ tenantId, consentedAt, consentVersion, ipAddress }`.

25.2 Data Deletion Workflow





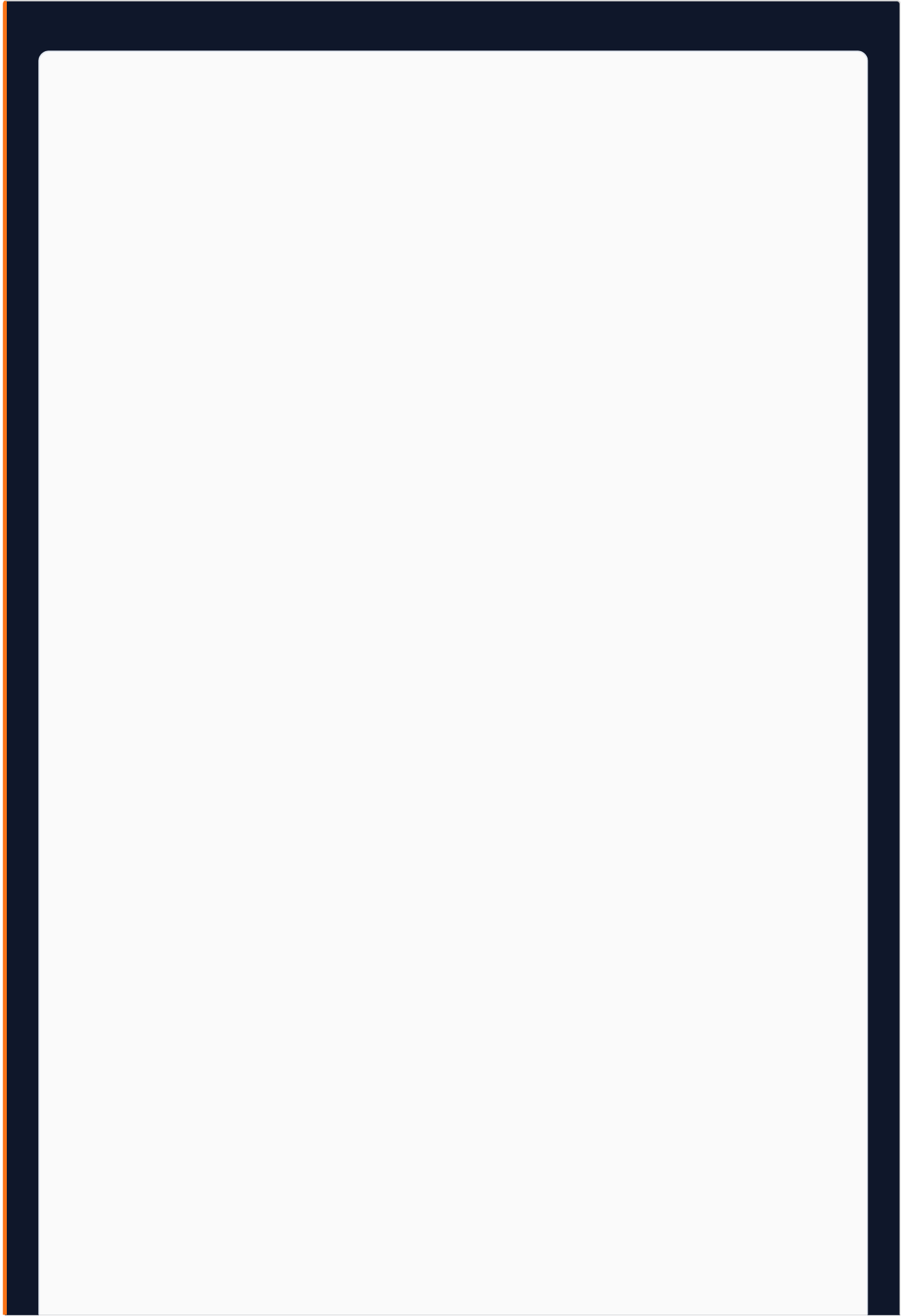


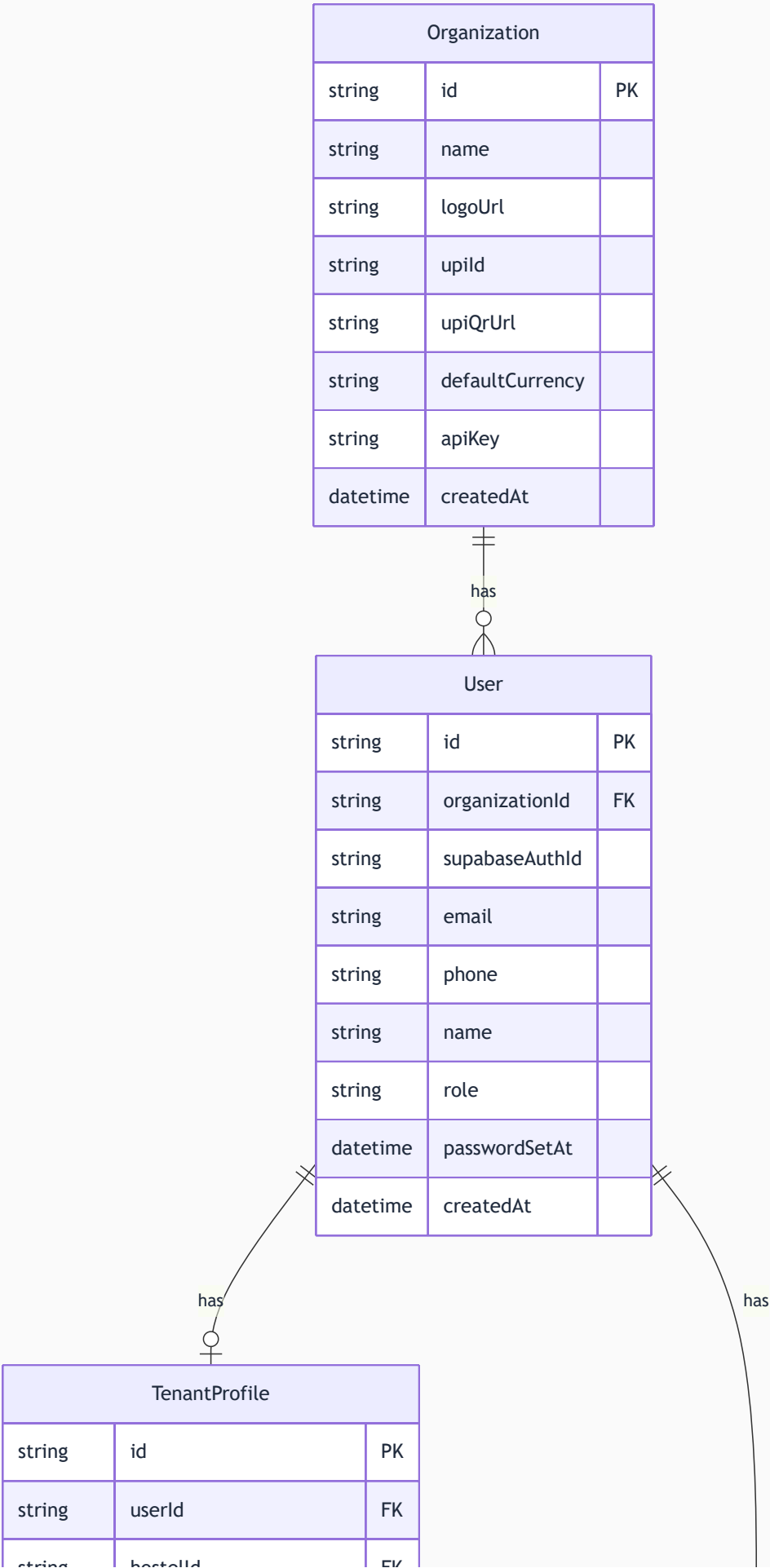
25.3 Data Retention Policy

DATA TYPE	RETENTION PERIOD	JUSTIFICATION
KYC Documents (S3 files)	Stay duration + 30 days	Deleted on DPDP request after stay
KYC Document Numbers	Stay duration + 30 days	Nulled on DPDP deletion
Payment Records	7 years	Income Tax Act - financial audit
Activity Logs	3 years	Internal audit and compliance
Visitor Logs	12 months	Security and law enforcement
Service Requests	2 years	Operational record keeping
Attendance Records	3 years	Labour law compliance
Food Transaction History	2 years	Financial audit trail

26. Data Architecture

| 26.1 Domain: Users and Authentication





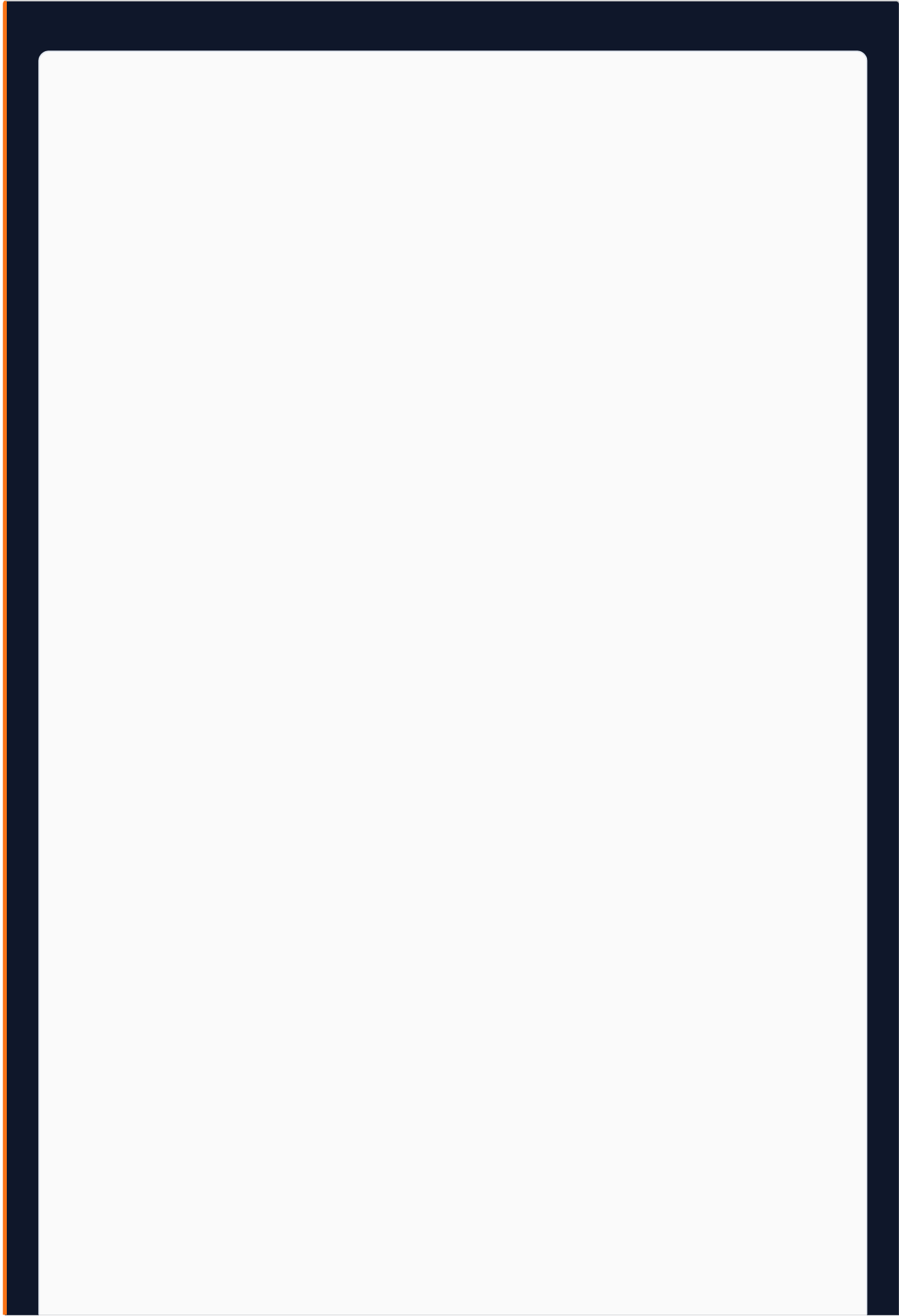
string	hostelId	FK
string	admissionId	
string	photoUrl	
string	aadhaarNumber	
string	panNumber	
string	licenseNumber	
string	kycDocumentType	
string	kycDocumentScanUrl	
string	emergencyContactName	
string	emergencyContactPhone	
boolean	dpdConsentGiven	
datetime	dpdConsentAt	
datetime	kycDeletedAt	
datetime	createdAt	

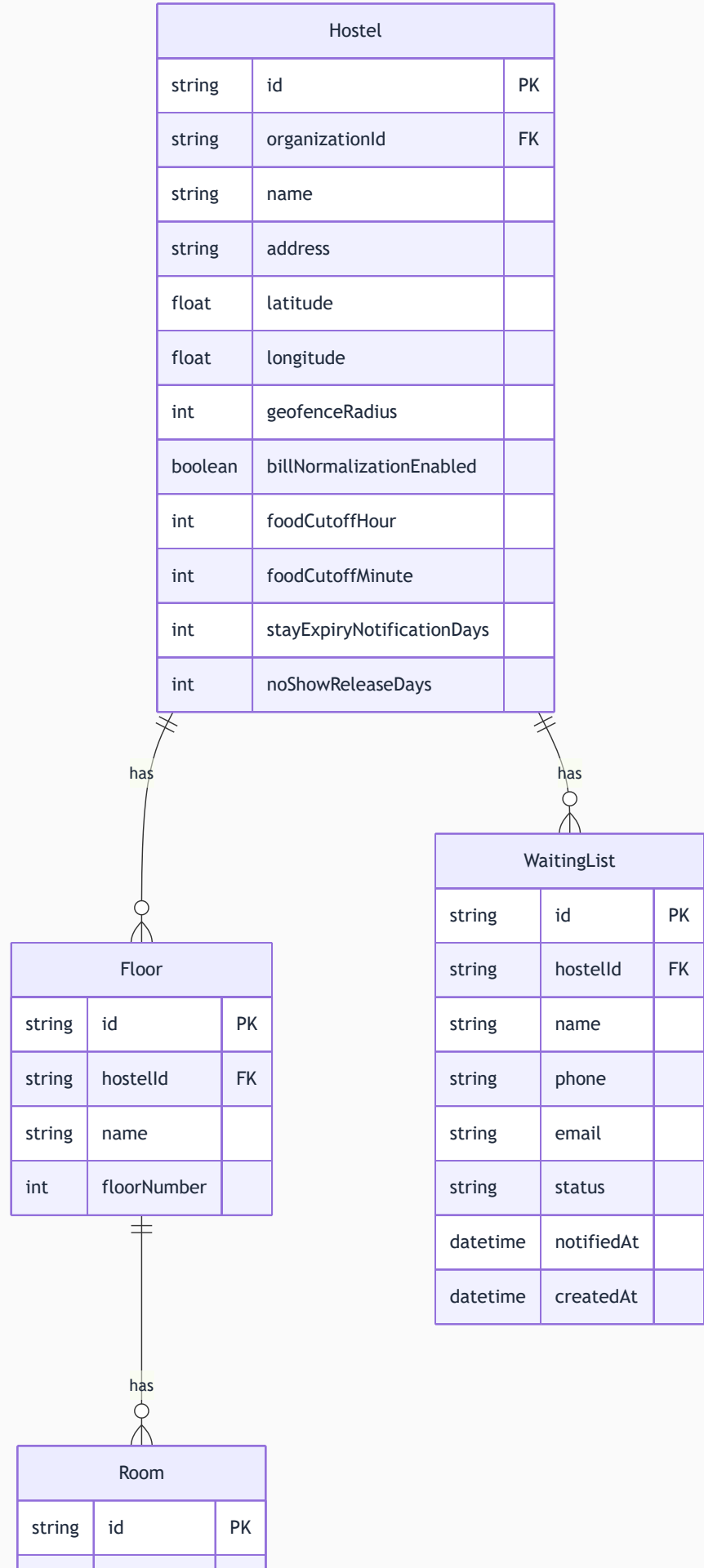


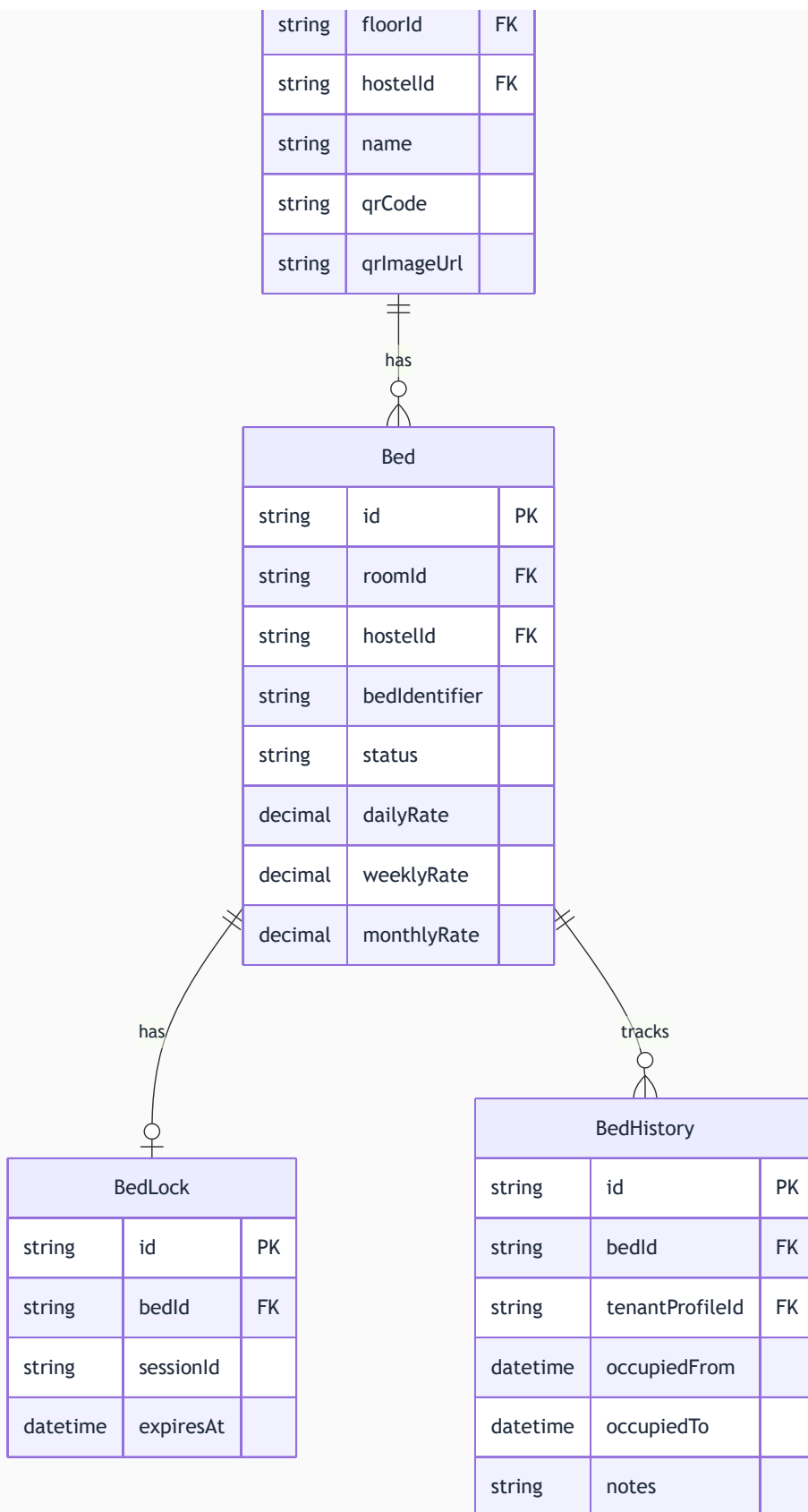
EmployeeProfile		
string	id	PK
string	userId	FK
string	hostelId	FK
string	designation	
string	workStartTime	
string	workEndTime	
datetime	joinedAt	



26.2 Domain: Hostel and Bed Hierarchy

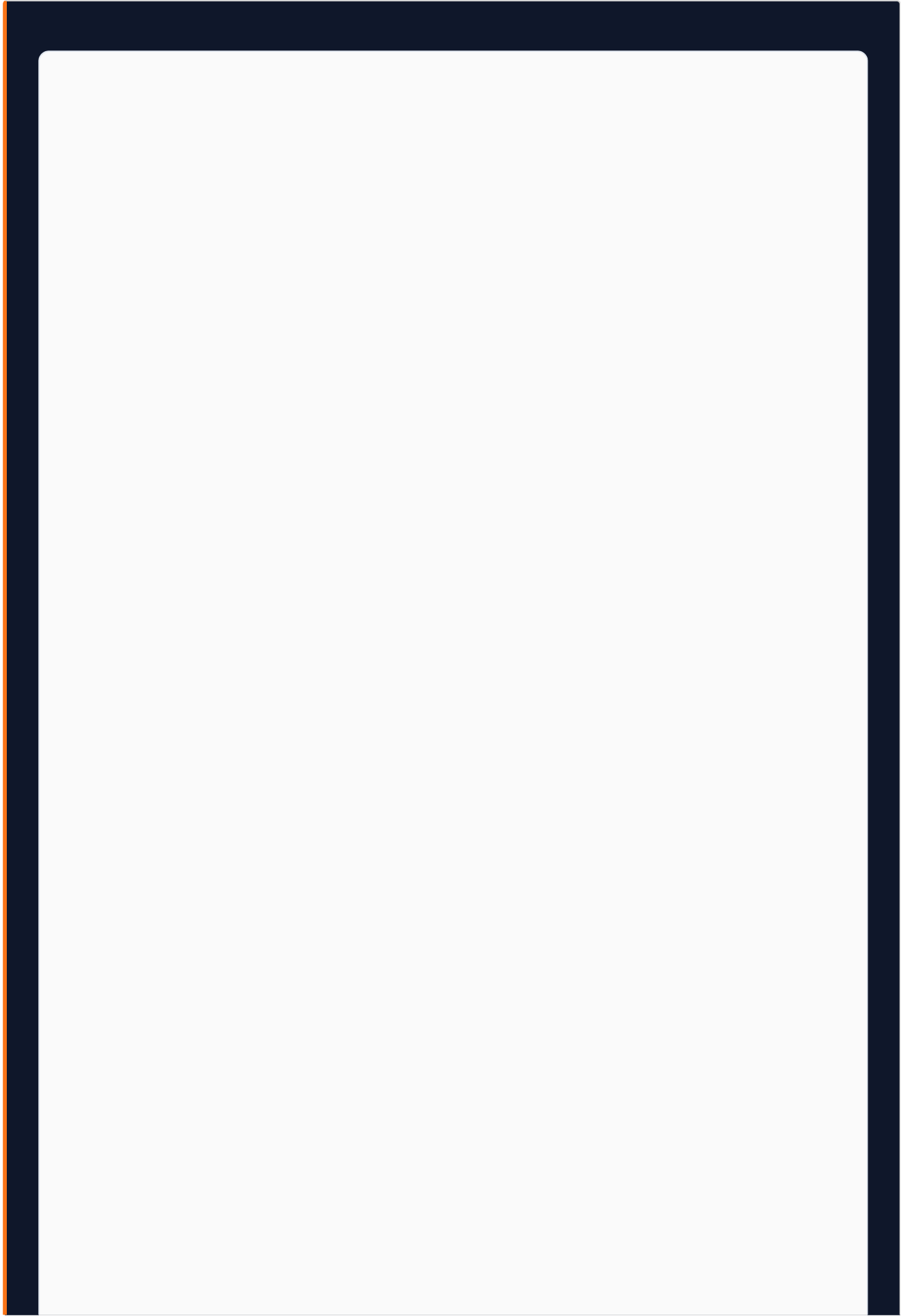


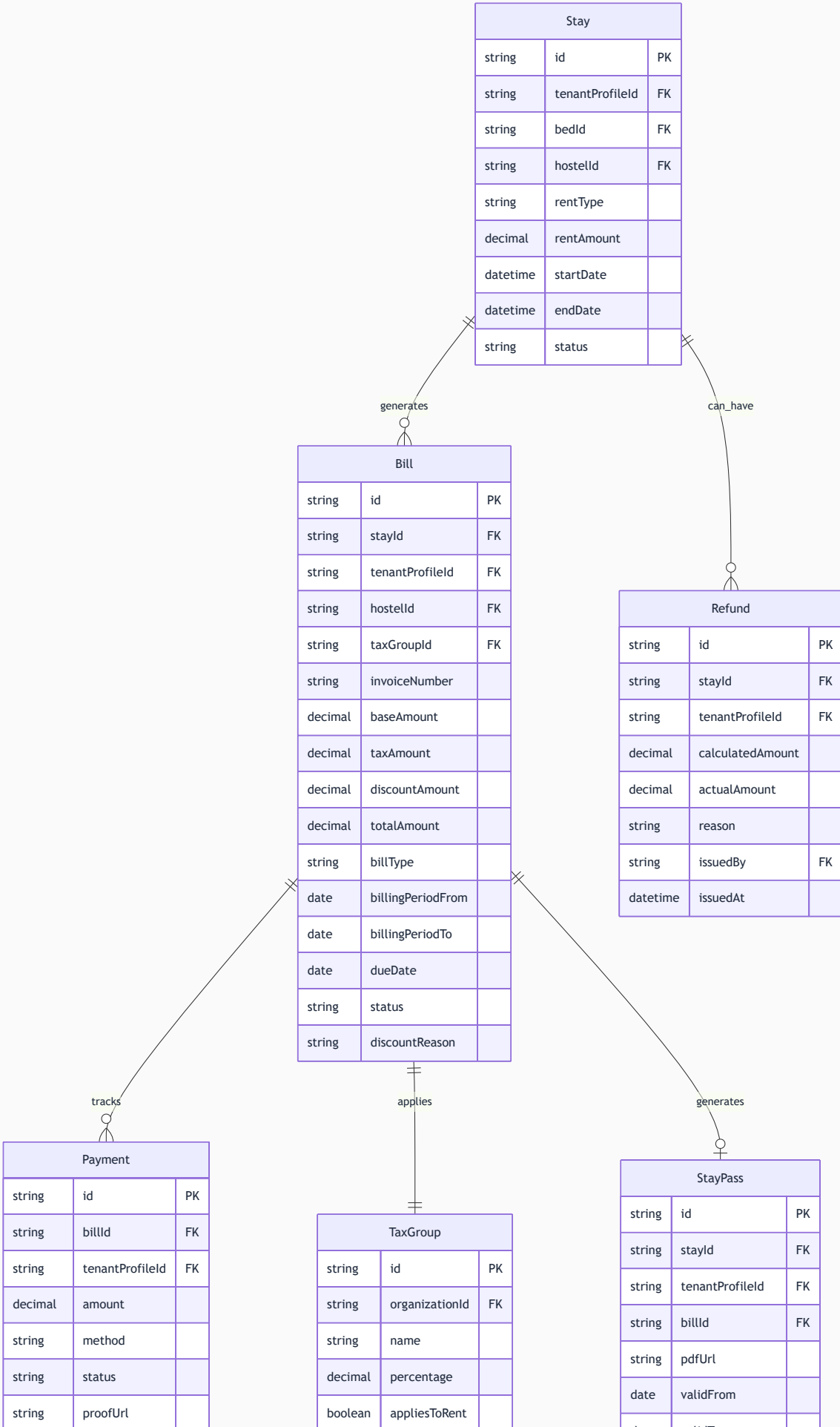






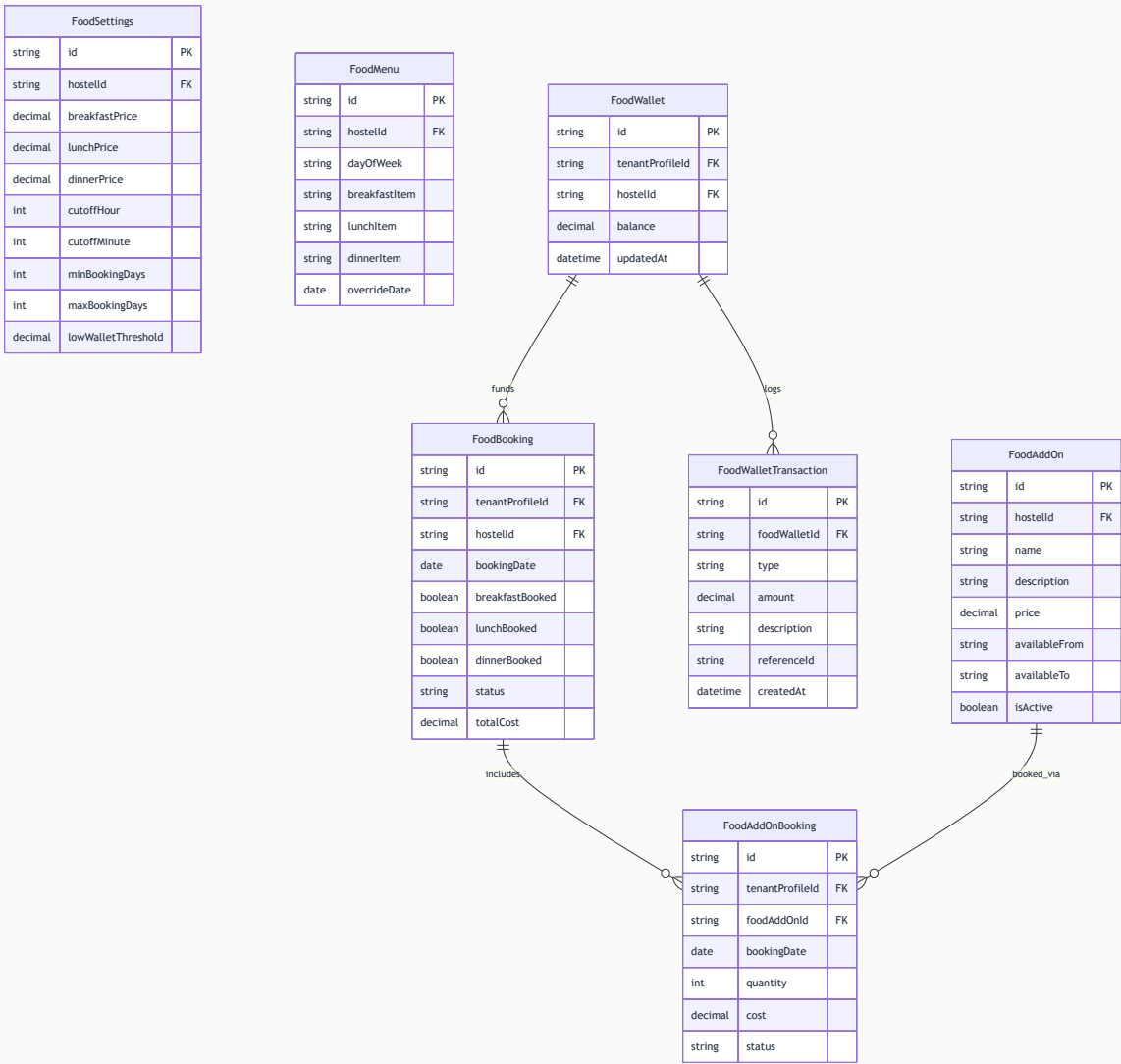
| 26.3 Domain: Stay and Billing



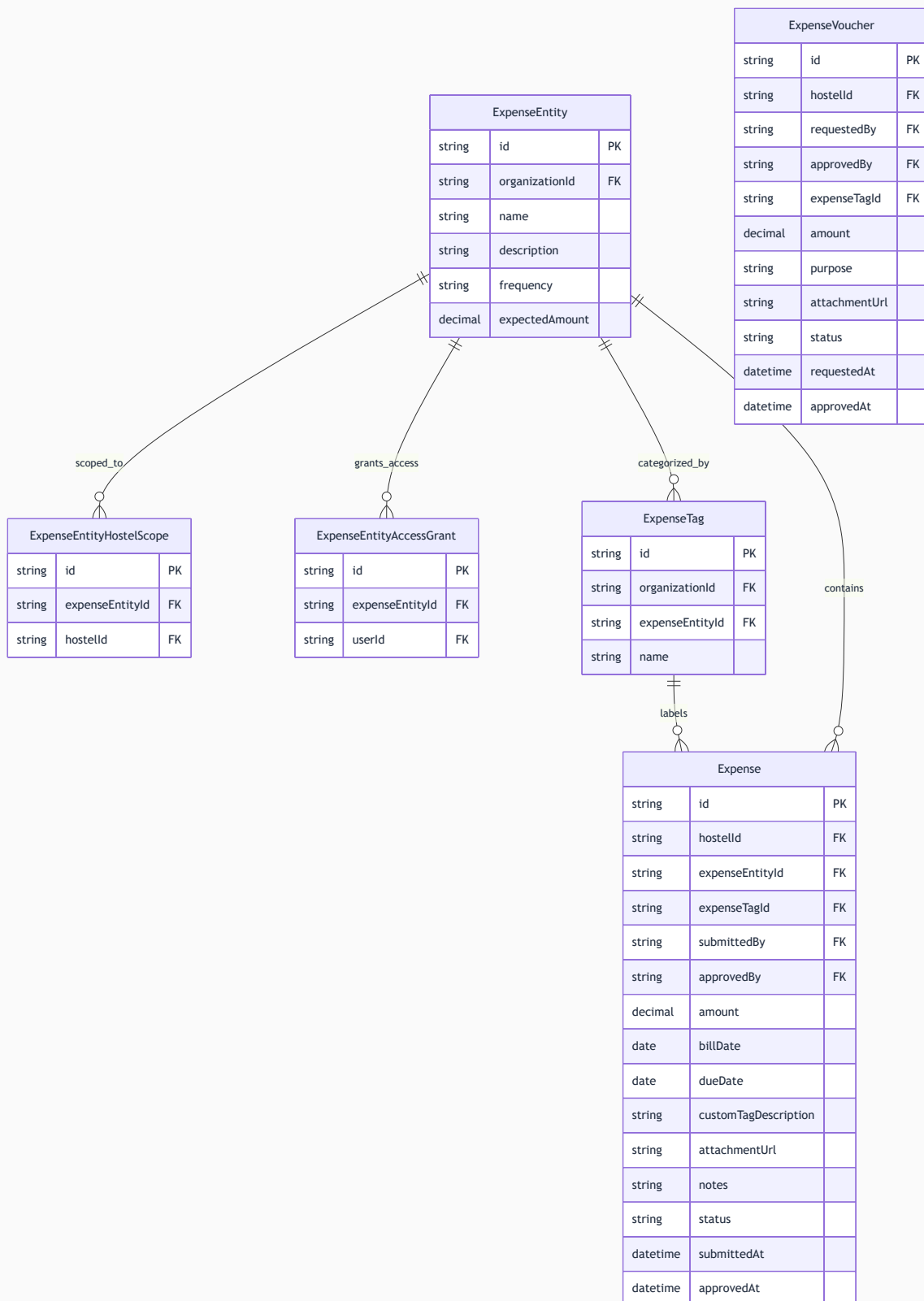


datetime	paidAt		boolean	appliesToFood		date	validTo	
string	verifiedBy	FK				string	status	
datetime	verifiedAt					string	verificationToken	

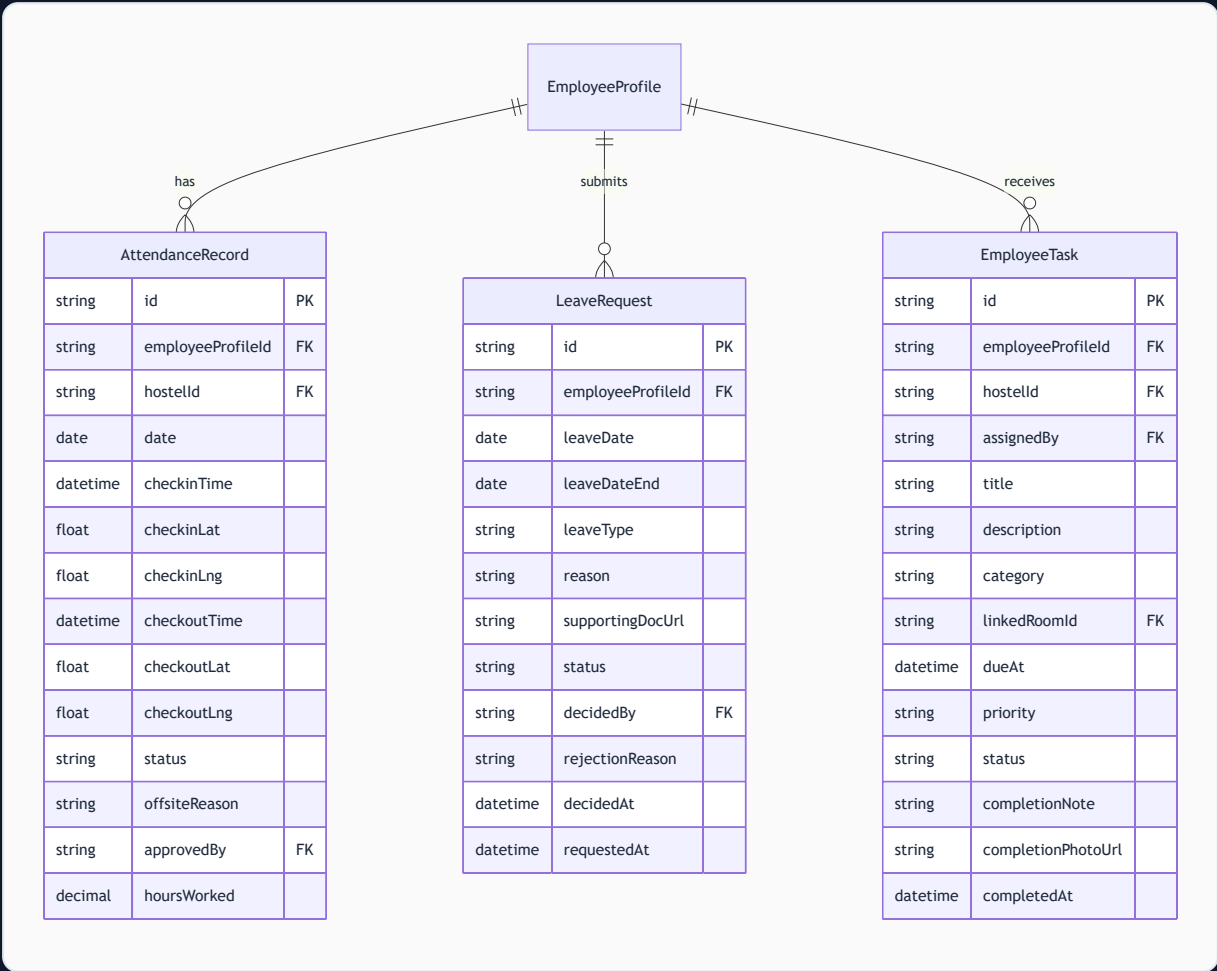
26.4 Domain: Food Management



26.5 Domain: Expense Management



26.6 Domain: Employee and Attendance



26.7 Domain: Service Requests, Visitors, and Notifications

ServiceRequest		
string	id	PK
string	hostelId	FK
string	submittedBy	FK
string	assignedTo	FK
string	category	
string	title	
string	description	
string	photoUrl	
string	priority	
string	status	
string	resolutionNotes	
datetime	resolvedAt	
datetime	createdAt	

VisitorLog		
string	id	PK
string	hostelId	FK
string	loggedBy	FK
string	tenantProfileId	FK
string	visitorName	
string	visitorPhone	
string	relationship	
string	purpose	
string	idProofType	
datetime	entryTime	
datetime	expectedExitTime	
datetime	actualExitTime	

Notification		
string	id	PK
string	userId	FK
string	organizationId	FK
string	type	
string	title	
string	body	
string	linkUrl	
boolean	isRead	
datetime	readAt	
datetime	createdAt	

Announcement		
string	id	PK
string	organizationId	FK
string	hostelId	FK
string	createdBy	FK
string	title	
string	body	
string	type	
date	expiryDate	
datetime	createdAt	

27. Infrastructure & AWS Architecture

27.1 V1 Infrastructure — Additions Over MVP

COMPONENT	MVP	V1 CHANGE
EC2 t3.micro	App server	Same — new features are additional API routes, not new services
RDS PostgreSQL t3.micro	Primary database	Schema significantly expanded — no tier change
S3	KYC documents	Expanded: PDFs, housekeeping photos, expense attachments, QR images, reports
CloudFront	Static assets	Same
Supabase Auth	Email/password	Extended with Google OAuth and Facebook OAuth
Google Maps API	Not used	Added for employee geolocation attendance

27.2 Background Jobs (Cron) — V1

All cron jobs run inside the Next.js process using node-cron. No separate infrastructure required.

JOB	SCHEDULE	PURPOSE
Bed Lock Cleanup	Every 5 minutes	Release expired BedLock records
Stay Expiry Checker	Daily at 8:00 AM	Identify expiring stays — fire notifications
No-Show Reservation Release	Daily at 9:00 AM	Auto-release reservations past no-show window
Food Cutoff Reminder	Daily at 9:00 PM	Notify tenants with unbooked meals for tomorrow
Food Wallet Low Balance Check	Daily at 8:00 AM	Notify tenants below wallet threshold
Late Fee Calculator	1st of month at 6:00 AM	Append late fees for normalized billing
Stay Pass Expiry Processor	Daily at midnight	Update expired pass statuses

27.3 S3 Bucket Structure

```
hostel-platform-prod/
├─ organizations/{orgId}/
│   └─ logo.png
│   └─ upi-qr.png
└─ hostels/{hostelId}/
    ├─ rooms/{roomId}/qr.png
    ├─ tenants/{tenantId}/
    │   └─ photo.jpg
    │   └─ kyc-document.jpg
    │   └─ documents/
    │       └─ stay-pass-{passId}.pdf
    │       └─ invoice-{billId}.pdf
    │       └─ receipt-{paymentId}.pdf
    └─ expenses/{expenseId}/attachment.pdf
    └─ housekeeping/{taskId}/
        └─ before.jpg
        └─ after.jpg
    └─ reports/{reportId}.pdf
```

27.4 Scaling Thresholds

TRIGGER	RECOMMENDED ACTION
EC2 CPU consistently above 70%	Upgrade to t3.small
RDS storage above 15 GB	Enable RDS storage autoscaling
30+ active hostels live	Evaluate ECS Fargate migration
S3 exceeding 5 GB	Standard S3 pricing kicks in
SWR polling load visible in CloudWatch	Begin SSE upgrade planning

28. V1.1 Roadmap — Deferred Features

28.1 Customizable Canvas Dashboard

What it is: Admins and wardens toggle an "Edit Dashboard" mode to drag, drop, resize, add, and remove widget tiles. Layouts persist per-user in the database.

Why deferred: A production-quality drag-and-drop widget system is a 3-4 week engineering effort. V1 ships with a carefully designed fixed layout that covers all operational needs.

V1.1 Technical Plan:

- Library: react-grid-layout
- Schema: DashboardLayout table with `{ userId, widgetType, x, y, w, h }` per widget
- Available widgets: Vacant Beds, Daily Revenue, Food Orders Today, Pending Payments, Active Service Requests, Complaint Count, Attendance Summary, Performance Score Badge

28.2 Regional Language Assistant — Malayalam

What it is: A toggle enabling contextual translation of specific UI elements (food menu items, room labels, hostel-specific terminology) into regional languages. Malayalam is first.

Why deferred: Correct contextual translation of custom hostel content cannot use Google Translate API at runtime — results are inaccurate for Indian food names and local terms. Requires a `Translation` database table where admins enter Malayalam equivalents of their custom content.

V1.1 Technical Plan:

- `TranslationEntry` table: `{ organizationId, entityType, entityId, language, translatedValue }`
 - Languages enum: MALAYALAM (expandable to Tamil, Telugu, Hindi, Kannada)
 - UI: Toggle in header, 'i' icon on translatable elements, admin translation management screen
-

29. MVP vs V1 Gap Analysis

CAPABILITY	MVP STATUS	V1 STATUS	EFFORT
QR-based self-service booking funnel	Not built	New	High
Bed hold mechanism with TTL	Not built	New	Medium
Visual room and bed selection UI	Not built	New	Medium
Group booking with linked accounts	Not built	New	Medium
Daily/Weekly rent types	Basic	Full engine	Low
Monthly rent with normalization	Basic	Full engine with normalization	High
Prorated first bill calculation	Not built	New	Medium
Tax groups on bills	Not built	New	Low
Discount application on bills	Not built	New	Low
Late payment fees	Not built	New	Medium
Advance rent refund workflow	Not built	New	Medium
Stay extension billing	Basic	Enhanced	Low
Next Month MRR Report	Not built	New	Medium
Finance Center and Bill Vault	Not built	New	Medium
Invoice PDF generation	Not built	New	Medium
Tenant payment receipt PDF	Not built	New	Low
Expense voucher (petty cash)	Not built	New	Low
Expense entities with access control	Not built	New	High
Expense tags system	Not built	New	Medium
Expense approval workflow	Not built	New	Medium
Expense attachment upload	Not built	New	Low
Per-meal food pricing model	Not built	New	High
Food wallet prepaid balance	Partial	Full system	Medium
Food calendar booking UI (tenant)	Basic	Enhanced with full cutoff logic	High
7-day minimum food booking rule	Not built	New	Medium
Admin-configurable food cutoff time	Not built	New	Low
Food cancellation within cutoff window	Not built	New	Medium
Daily food procurement report	Not built	New	Medium

CAPABILITY	MVP STATUS	V1 STATUS	EFFORT
Weekly food menu management	Partial	Enhanced	Low
Food add-on items	Not built	New	Medium
Bed Transfer workflow	Not built	New	Medium
Bed Swap workflow (atomic)	Not built	New	Low
Bed Reservation for future date	Not built	New	Medium
Waiting list for full hostels	Not built	New	Low
Stay Pass PDF generation	Not built	New	Medium
Stay Pass QR verification page	Not built	New	Low
Stay Pass auto-expiry lifecycle	Not built	New	Medium
Lease renewal self-service flow	Basic	Full workflow	Medium
Overstay detection and alerting	Not built	New	Low
Unified service request module	Ticketing partial	Full - 3 categories	Medium
Service request SLA tracking	Not built	New	Low
Visitor and Guest log	Not built	New	Low
Employee user type and portal	Not built	New	High
Geolocation attendance marking	Not built	New	High
Out-of-premises attendance approval	Not built	New	Medium
Employee leave request system	Not built	New	Medium
Employee task assignment (enhanced)	Partial	Enhanced	Low
Housekeeping scheduling module	Not built	New	Medium
Housekeeping photo verification	Not built	New	Low
Hostel performance scoring engine	Not built	New	High
Comparisons and benchmarking module	Not built	New	High
PDF comparison report generation	Not built	New	Medium
Announcements module (enhanced)	Partial	Enhanced	Low
Asset and inventory management	Not built	New	Medium
Currency display switch (INR to AED)	Not built	New	Low
Google OAuth for tenants	Not built	New	Medium
Facebook OAuth for tenants	Not built	New	Medium
DPDP consent collection at onboarding	Not built	New	Low
KYC document deletion workflow	Not built	New	Low
Room QR code generation and download	Not built	New	Low

CAPABILITY	MVP STATUS	V1 STATUS	EFFORT
Admin dashboard full Pinpoint redesign	Basic	Full redesign	High
Hostel performance dashboard cards	Basic	Full scoring cards	High
Integration-ready API key schema	Not built	Schema and design only	Low
Webhook event architecture (foundation)	Not built	Schema and design only	Low

Appendix A: Glossary

TERM	DEFINITION
HOS	Hostel Operating System — the V1 platform positioning
Normalization	Aligning all monthly tenant billing cycles to the 1st of the calendar month
Bed Hold	A time-limited (10-minute TTL) database lock placed on a bed while a prospective tenant fills the booking form
Stay Pass	The official digital credential (PDF) confirming a tenant's right to reside in the hostel
Admission ID	A unique sequential identifier assigned to each tenant at the time of onboarding approval
Bill Vault	The searchable archive of all financial documents (invoices, receipts, vouchers) stored in S3
Procurement Report	A daily or weekly count of food portions needed by the kitchen for the next day or week
Food Wallet	A prepaid balance maintained by each tenant to fund their meal bookings
Expense Entity	A named, recurring expense obligation (e.g., "Building Rent — NHP") with access control
Expense Tag	A sub-category label within an expense entity for granular categorization
Expense Voucher	A warden-initiated petty cash request for on-premises operational expenses
MRR	Monthly Recurring Revenue — the forecasted revenue for the next calendar month
Geofence	A virtual geographic perimeter around the hostel location for employee attendance proximity verification
DPDP	Digital Personal Data Protection Act 2023 (India)
SSE	Server-Sent Events — the planned Phase 2 real-time data upgrade deferred from V1
SWR	Stale-While-Revalidate — the data fetching strategy used for real-time dashboard polling
TTL	Time-To-Live — the expiry duration on bed locks and other temporary records
BedLock	The database record implementing a bed hold with a TTL expiry timestamp
Stay Extension	Adding additional days, weeks, or months to an existing stay
Renewal	A formal restart of a new billing cycle, typically month-to-month
Waiting List	A FIFO queue of prospective tenants who registered interest when the hostel had no available beds

Appendix B: Open Items & Pending Decisions

ITEM	STATUS	OWNER
Platform branding — "Stayee" vs "Anywhere Node" vs other name	Pending co-founder confirmation	Founders
Hostel scoring algorithm weight calibration	Weights defined — validate with first 2-3 pilot hostels	Product
Employee leave types annual count and carry-forward rules	Casual/Sick/Emergency defined — leave count and carryover TBD	Product
Visitor log export format for law enforcement compliance	Confirm acceptable format with legal review	Legal / Compliance
WhatsApp notification integration	Deferred — evaluate for V1.1	Engineering
Multi-hostel employee edge cases	Basic support defined — full multi-hostel schedule management TBD	Engineering
Super Admin portal feature scope	Minimal V1 scope — expand in V2	Product
Tally / Zoho Books integration	API architecture is integration-ready — actual integration deferred	Future
Employee salary and payroll module	Explicitly out of V1 and V1.1 scope	Future
360-degree room view for booking funnel	Explicitly deferred to V3	Future
Room photos in booking funnel	Explicitly deferred to V3	Future

This document is version 1.0. It will be updated as decisions are finalized and engineering assessments are completed.

Next step: Review and approval of this Concept Document, followed by authoring the Product Requirements Document (PRD) with detailed acceptance criteria per module.

End of Document

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